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Explainable framework based on integrated sensor fusion and deep learning to estimate stability in high-speed micromilling of thin-walled titanium alloy

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ABSTRACT

A novel deep learning (DL) framework based on multi-sensor fusion based on vibration and audio signals with transfer learning has been developed and validated for chatter onset prediction in micromilling of thin-walled titanium (TC4) alloys. The DL framework has been trained and validated with individual sensor data, labelled using machined surface micrographs and cutting tool condition. A total of 879,840 spectrograms has been generated for training via signal segmentation and augmentation methods that yielded a validation accuracy of 89 % during training phase. The trained DL model has been tested on the unlabelled fused signal data to make the DL model adapt to the unseen patterns in the signals recorded during real-time machining. The proposed DL framework based on fused signals outperformed individual sensor-based detection methods by demonstrating an increment of 33 % in accuracy, with a mean absolute error of 0.027416 and R^2 of 0.987. Ablation studies confirm that fusing signals enhances the DL model's performance, yielding a 21 % and 29 % improvement in precision and recall, respectively. The optimal fusion strategy reduced chatter misclassification, with one instance of misclassification. The proposed framework presents a robust, explainable DL approach that can be readily used for adaptive control of cutting conditions in different manufacturing processes. The industry readiness of the DL model has been demonstrated by testing the pretrained model on turning process. The DL model shows 95.31 % prediction accuracy with only three misclassifications out of sixty-four test conditions.

1. Introduction

Mechanical Micromilling is the most preferred process for fabricating thin-walled structures like blisks, ribs, frames, turbine blades, window panels, and stents, etc., for variety of industrial applications (Herranz et al., 2005). For these components, the feature size goes as low as 50 μm and requires a cutting tool of 10–30 μm in diameter. At these low sizes, chatter is inevitable as the stiffness reduces with every pass of the cutting tool over the flexible workpiece. Early chatter onset detection is highly useful in reducing the detrimental effects of chatter like dimensional losses with poor surface finish (Brajesh et al., 2022). Also, these detection techniques can be used as feedback for real-time monitoring of the cutting process. Different methods that exist to control the losses are spindle speed modification (Brajesh et al., 2022), micro-endmill tool path optimization (Novakov and Jackson, 2010), and damping techniques (Jin and Altintas, 2013; Tuysuz and Altintas, 2019). Conventional approaches like solving the governing equation and generating the stability lobes for chatter free cutting parameters rely more on experimentally predicted cutting coefficients and micro-end mill dynamics and moreover, these predicted parameters are material dependent (Heitz et al., 2022; Serje et al., 2020) and cutting conditions dependent. Hence, this requires a more generic process, which is independent of the tool-work combination and provides an accurate and reliable chatter onset prediction. Over the past decade, Machine Learning (ML) has proved to be a possible solution to the problem, but the requirement of large data sets to train the model, hyperparameter tuning, extensive hardware requirements are the hinderances to its applicability. Reported literatures show extensive work on reducing the computational complexities to make ML-based solution as feasible and reliable (Nasir and Sassani, 2021; Soori et al., 2023). The approaches involved developing new algorithms and/or modifying the existing algorithms for image processing, domain conversion of signals, IoT devices, MEMS/NEMS devices, etc. But, for real-time process monitoring of thin-walled TC4 alloys, the chatter onset should be predicted as early

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as possible so that the process parameters can be altered instantaneously. This does not require a customised or advanced algorithms with large computational resources. In the present work, it is shown that a feed forward CNN with back propagation can predict the chatter onset with 96.87 % accuracy. A significant improvement in prediction accuracy and F1 score in comparison to single-sensor approaches is observed. The proposed and validated DL model can be used as feedback to predict the machining stability without requiring complex signal transformations or solving governing equations, making it computationally efficient and reliable solution for industrial applications. Furthermore, the trained DL model has also been tested for stability estimation of the turning process at different cutting conditions. The DL model demonstrated a prediction accuracy of 95.31 % with only three misclassifications out of sixty-four test conditions, validated by machined surface analysis. The enhanced adaptability of the proposed DL model for the unlabelled data shows the transfer learning capability, highlighting the industry readiness of the model.

2. State of the art about the chatter identification strategies

2.1. Existing chatter identification strategies

Chatter detection in micro-milling has been studied extensively using several analytical techniques that focus on time and frequency domain vibration signal analysis. A frequently based approach involves using a dimensionless chatter indicator based on root-mean-square (RMS) acceleration values calculated per spindle revolution. While effective, these methods face limitations with low prediction accuracy as they fail to consider the dynamic interactions between the tool and workpiece, particularly the size effect. More sophisticated techniques, such as wavelet packet transform (WPT), provide deeper insights into the dynamics but are impractical for real-time applications due to their computational complexity. Chatter onset is typically associated with a significant increase in vibration amplitude and a shift in dominant frequency components, which complicates the prediction of machining stability in real-time monitoring (Li et al., 2014). The two main strategies for chatter identification include regenerative delay and parametric forcing. Regenerative delay arises from the wavy surface left behind after previous cutting passes, while parametric forcing is caused by periodic excitation of the cutting tool. These phenomena are modelled using delay-differential equations with time-periodic coefficients, but the inherent complexity of these models limits their utility for real-time monitoring. Methods such as semi-discretization have been developed to improve prediction accuracy; however, reliable forecasting of chatter remains difficult, particularly under different machining conditions, due to the intricate dynamic interactions within the system (Insperger et al., 2015). Stability lobe diagrams (SLDs) are often employed to predict chatter onset, but their accuracy is limited by challenges in estimation of cutting coefficients that affects surface integrity and tool life (BAI et al., 2024).

Displacement sensors, average power spectral density (APSD) analysis and surface topography analysis have proven effective in capturing chatter onset (Singh et al., 2015). Traditional force sensors, on the other hand, have been found inadequate for high-speed micromilling of thin-walled structures due to their limited frequency bandwidth. Surface topography analysis is a reliable method for chatter detection, but removal and replacement of workpiece adds geometrical errors. Additionally, in-process chatter detection strategies are proposed that utilizes cutting force signals, vibration signals, displacement signals, laser signals and current signals, etc., are processed through variational mode decomposition (VMD) and multi-scale permutation entropy (MSPE), WPTs, etc. These approaches can be used to classify stable, mild, and severe chatter conditions using support vector machines (SVM), Neural networks, DL, etc., that can achieve 100 % classification accuracy outperforming other methods, such as empirical mode decomposition (EEMD), which achieved lower accuracy (Wang et al., 2023). Although these advanced techniques represent significant progress in chatter detection as post-processing methods but fail in real-time monitoring, as these methods cannot generalize the learning gained in training to the new unseen data recorded in real-time. (Wang et al., 2022a; Sestito et al., 2022; Navarro-Devia et al., 2023).

2.2. Multi-sensor fusion strategies

Multi-sensor fusion has significantly improved chatter detection accuracy in milling. Early methods integrated axial force sensors and accelerometers, addressing limitations like narrow frequency bandwidths and environmental sensitivity. Despite this, challenges in noise suppression and distinguishing chatter from ambient noise, required further improvements in the fusion strategies (Kuljanic et al., 2008). Later advancements combined accelerometers and force sensors with neural networks and advanced signal processing techniques. Although these systems improved performance, issues with sensor placement, environmental noise and occasional sensor malfunctions hindered practical application (Kuljanic et al., 2009). Artificial intelligence approaches, which focused on analyzing vibratory signals at resonance frequencies, further enhanced detection accuracy. However, the complexity of feature selection and computational demands limited their scalability for real-time use (Lamraoui et al., 2015). Methods such as the fusion of Variational Mode Decomposition (VMD) and Wavelet Packet Decomposition (WPD) improved early chatter detection through energy entropy analysis, but real-time implementation remained challenging due to the complexity of selecting wavelet bases (Zhang et al., 2016). The integration of vibration signal features with neural networks, such as self-organizing maps (SOM), helped in detecting chatter before significant workpiece damage. However, high computational demands restricted real-time application (Cao et al., 2017).

Ensemble Empirical Mode Decomposition (EEMD) was used to analyze multi-channel vibration signals with Fisher discriminant ratios for feature selection, achieving high classification accuracy with a two-channel strategy. However, real-time application faced limitations due to the need for precise channel selection and significant computational intensity (Chen et al., 2018). Other strategies, such as using noise reduction and feature extraction with adaptive network-based fuzzy inference systems (ANFIS), improved predictions of tool life. Despite enhancing accuracy, the complexity of real-time processing and system tuning made industrial deployment difficult (Wu et al., 2018). In wireless sensor networks, Multi-sensor fusion improved energy efficiency and data accuracy through frequency- and range-based fusion strategies. Although accuracy and efficiency were enhanced, challenges remained in managing overlapping data and ensuring real-time processing (de Farias et al., 2019). Advanced methods, incorporating time and frequency domain features with machine learning models like stacked-denoising autoencoders (SDAE) and Adaboost-SVM classifiers, demonstrated high classification accuracy, but the computational demands limited their real-time scalability (Wan et al., 2021). Recent research improved fusion strategies using upgraded Principal Component Analysis (PCA) and energy-based feature selection, achieving 93.75 % detection accuracy. However, computational complexity remained a significant barrier to real-time application (Wang et al., 2022b). Other approaches combined Multi-sensor signal fusion with multiscale entropy analysis to improve online chatter detection accuracy, though the complexity of the fusion algorithm and logistic regression classifier limits industrial use (Liu et al., 2023). The

Dual-Stream Residual Attention CNN method optimized feature extraction using acceleration and cutting force signals, achieving 98.68 % detection accuracy, but the computational demands restricted real-time deployment (Zhan et al., 2024).

2.3. Deep learning strategy

An adaptive data fusion strategy adopting CNN was implemented to improve prediction accuracy by fusing data from multiple sensors. The proposed approach employed a 1D CNN for feature extraction with batch normalization for enhanced prediction capability. Despite improved prediction accuracy, real-time diagnostics encountered computational challenges due to dilated convolution (Li et al., 2020). In-process tool condition forecasting applied a deep learning framework combining LSTM and residual CNNs to predict tool wear. By capturing historical data and predicting future values, the model offered real-time updates. While demonstrating strong predictive capabilities, the method's dependency on large datasets and the LSTM's complexity presented challenges, particularly for real-time implementation (Sun et al., 2020). Milling chatter detection leveraged scalogram images of cutting force signals, which were classified using CNNs. This method achieved a 99.67 % classification accuracy, surpassing traditional techniques. However, the computational cost and memory requirements were substantial due to high-resolution input data and reliance on powerful GPUs for training (Tran et al., 2020). A method combining convolutional and recurrent neural networks was developed for online part deformation prediction during machining, eliminating the need for residual stress measurements. It improved prediction accuracy with errors reduced to 0.029 mm. However, challenges persisted in generalizing the model to different machining materials and processes (Zhao et al., 2020). A multi-domain feature fusion strategy using CNNs was introduced for tool wear prediction in milling operations. By analyzing vibration and cutting force signals, the model demonstrated superior performance to traditional approaches. Despite its success, the method required extensive preprocessing and labelled data, with noisy data posing accuracy challenges (Huang et al., 2020). Machine state monitoring in precision machining utilized audio and vibration signals for fault detection and operational setting prediction. Fast Fourier Transform and Principal Component Analysis were applied, and classifiers like KNN (Tahri et al., 2016), CNN (Harrou et al., 2020), Transformers (Hittawe et al., 2024) and SVM (Hittawe et al., 2015) achieved high classification accuracy. However, the need for generalization across different machining setups presented challenges for real-world application (Han et al., 2021).

2.4. Identified gap and proposed novel approach

In sections 2.1, 2.2, and 2.3, it is emphasized that chatter is an unwanted companion in micromilling. Literature[s] has shown that traditional methods for chatter detection often fail to accurately predict the onset of chatter in real-time monitoring. Because these methods rely on predicted material and process-specific cutting coefficients and dynamics and complex signal transformations, which are time-consuming for real-time monitoring. To overcome these limitations, the present work proposes:

1. A deep learning approach based on multi-sensor fusion: Fusing vibration and audio signals for reliable chatter onset identification.
2. VMD is used to reconstruct the raw vibration and audio signals by selecting specific components of the signal that contains crucial information required for the chatter identification.
3. Two-direction vibration (feed and normal-to-feed) signal and audio signal from the microphone at 60° (Gururaja and Singh, 2023) are fused together using different fusion strategies like correlation, weighted average, mean, median, etc.
4. Fused signal spectrograms, generated via short-time Fourier transforms (STFTs) are used to train the DL model and the performance of the DL model has been evaluated and validated with the machined surface analysis.
5. The optimal performance metric for the DL model and the optimal fusion strategy are estimated using the explainable deep learning (xDL) analysis techniques and ranking methods, respectively.
6. The optimal fusion strategy embedded DL model has been used to identify chatter onset in turning operation of Ferritic Stainless Steel, FSS-409 to evaluate the DL model's adaptability to the new, unlabelled and unseen data from different machining environment. This highlights the DL model as a single solution for stability estimation in multiple manufacturing processes.

3. Methodology and experimentation

3.1. Proposed chatter identification strategy

Over the past 4-5 decades signals from the contact-based sensors are the most used for chatter identification (Gururaja and Singh, 2024). Physically, the accelerometer captures direct vibrations transmitted through the tool and workpiece, while the microphone detects indirect emissions (airborne sound) produced during the cutting process. However, contact-based sensors, which are used to capture cutting dynamics, are constrained by limited bandwidth (Singh et al., 2015), and non-contact sensors are highly sensitive to acoustic noise from the surrounding environment. As a result, signal filtering often leads to the loss of critical low-amplitude chatter information. The present work proposes fusion of vibration and audio signals as the accelerometer captures the low-to mid-frequency vibration components while the microphone detects high-frequency acoustic emissions. The fused signal allowed for smoother data with minimal noise distortions, as filtering is restricted to basic notch filtering to remove transmission noise. Furthermore, for rapid chatter onset detection, identification and isolation of the frequency components contained within the signal is essential. Although wavelet transforms are effective at capturing time-localized features of a signal, they tend to suffer from poor frequency resolution in low-frequency ranges. Hence, in the present work, to extract the desired frequency components from the signal, VMD is used. VMD provides a higher frequency resolution by iteratively decomposing the signal into narrow-band modes, allowing for precise identification of specific frequency components, which is crucial for understanding the dynamics in micromilling. After applying VMD, different signal fusion strategies have been implemented to combine the vibration and audio signals. Spectrograms are then generated using STFTs, which are utilized to train, test, and validate the DL model. For model training, the signals are segmented using a windowing operation, with spectrograms generated for each segment and for testing the model, no signal segmentation is used because the trained DL model will be used for real-time monitoring. The optimal performance metric of the proposed DL model for chatter onset detection has been evaluated using the partial dependence (PDP) and individual conditional expectation (ICE) plots. Additionally, the proposed fusion strategies have been ranked based on error analysis, with R^2, MAE, and Explained variance as the ranking parameters. Based on this ranking and the PDP-ICE analysis, the optimal fusion strategy is selected. For the DL model with the identified optimal fusion strategy, ablation study and spectral analysis have also been conducted to examine the effect of signal fusion on the DL model's performance. Fig. 1

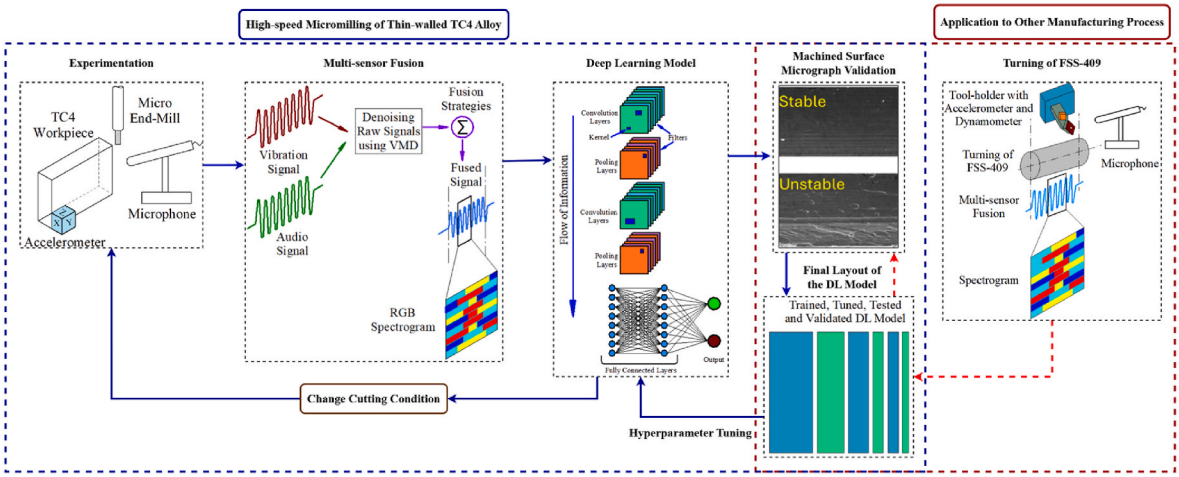


Fig. 1. Flowchart of the adopted methodology.

provides the flow of the proposed multi-sensor fusion integrated DL strategy for chatter identification in different manufacturing environments. Furthermore, machined surface analysis has also been conducted to validate the results, where both outcomes complement each other. Also, the pretrained model has been used to identify the chatter onset in the turning process also as shown in Fig. 1 to demonstrate the transfer learning capability of the DL model. The pseudocode for the entire approach is illustrated in Appendix-A.

3.2. Experimental setup and cutting conditions

High-speed low immersion radial micromilling experiments are performed on the thin-walled TC4 workpiece using micromachining. The three-axes of the micromachining centre has linear encoders with 0.15 μm resolution and an accuracy of ±2 μm for all the three linear axes. The spindle of the machining centre can reach a maximum of 80000 rpm, and it is cooled by an air-oil mixture with a maximum torque of 11.9 N-cm. The CAD model of the machining centre with arrangement of sensors and thin-walled TC4 workpiece is shown in Fig. 2(a). Fig. 2(b) shows the machining area that encompass the arrangement for accel-

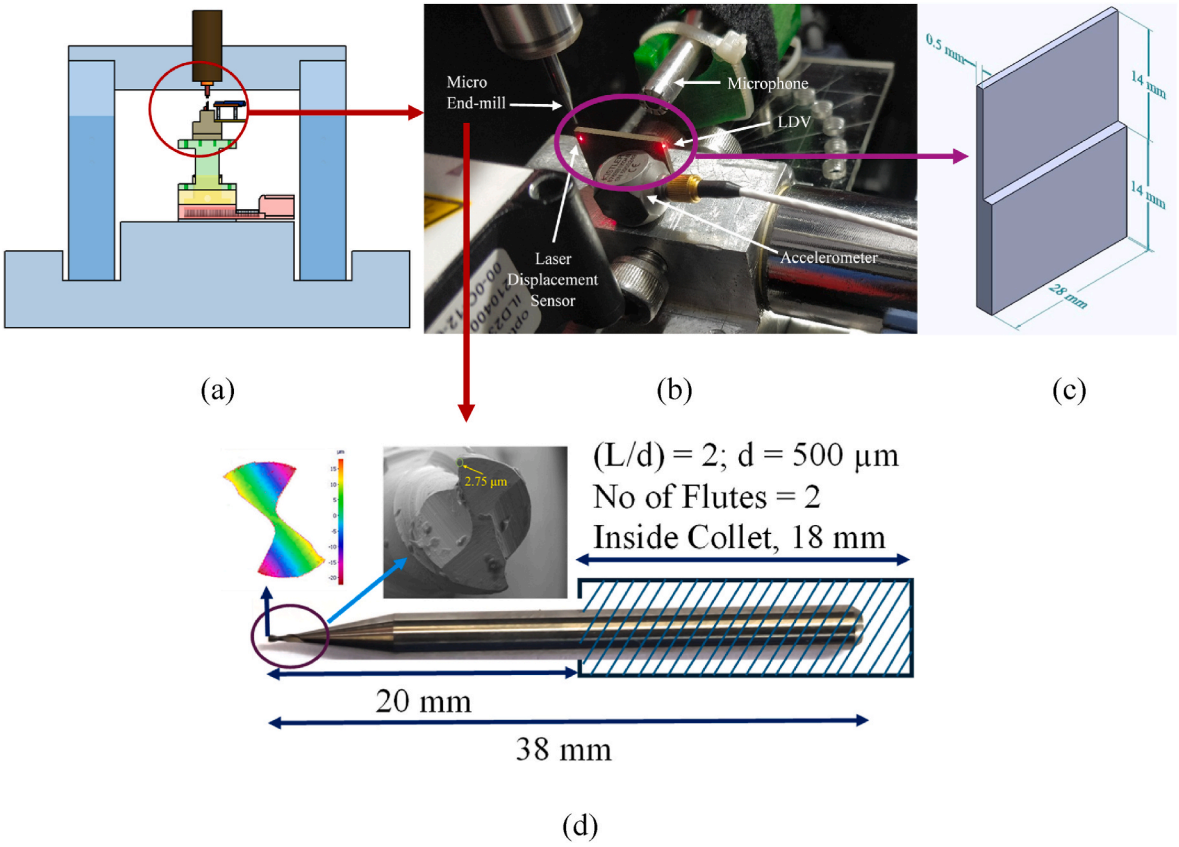


Fig. 2. Experimental setup; (a) High-speed machining centre; (b) Machining zone with all sensors; (c) Thin-walled TC4 workpiece; (d) Micro-end mill configuration.

Table 1
Radial micromilling cutting conditions.

Cutting Condition	Spindle Speed rpm	Radial Depth of Cut μm	Feed Rate mm/min	Cutting Condition	Spindle Speed rpm	Radial Depth of Cut μm	Feed Rate mm/min
1	20000	10	120	17	20000	50	120
2	20000	10	200	18	20000	50	200
3	40000	10	240	19	40000	50	240
4	60000	10	360	20	60000	50	360
5	40000	10	400	21	40000	50	400
6	80000	10	480	22	80000	50	480
7	60000	10	600	23	60000	50	600
8	80000	10	800	24	80000	50	800
9	20000	30	120	25	20000	70	120
10	20000	30	200	26	20000	70	200
11	40000	30	240	27	40000	70	240
12	60000	30	360	28	60000	70	360
13	40000	30	400	29	40000	70	400
14	80000	30	480	30	80000	70	480
15	60000	30	600	31	60000	70	600
16	80000	30	800	32	80000	70	800

erometer, microphone, and displacement sensors to capture the cutting dynamics and the CAD model of the thin-walled TC4 workpiece is shown in Fig. 2(c). All the low immersion radial micromilling experiments, are conducted in dry condition using the uncoated tungsten carbide end mill (see Fig. 2(d)). The micro-end mill has an edge radius of 2.75 μm, as measured from a SEM image of the micro-end mill (Fig. 2(d)). The overhang length of the cutting tool is kept at 52 % of the overall length for all the experiments as shown in Fig. 2(d).

The full factorial design of experiments contains four levels of spindle speed (20000 rpm–80000 rpm in intervals of 20000 rpm), a constant axial depth of 50 μm, two levels of feed rate (3 μm/flute and 5 μm/flute) and four levels of radial depth of cut (30 μm–180 μm in intervals of 50 μm) which accounts to a total of thirty-two experiments as shown in Table 1. All the experiments have been carried out using a free-field microphone (PCB, 130F20) with 38.7 mV/Pa sensitivity stationed at a tilt angle of 60° and at a stand-off distance of 25 mm (Gururaja and Singh, 2023). The vibration of the thin-walled workpiece while machining is recorded by a tri-axial accelerometer (Kistler, 8763B1K0AB, ±1000g). The displacement of the workpiece in normal to cutting direction is recorded by a laser sensor (OptoNCDT, ILD-2300-2). The workpiece vibration along the machining direction is recorded using a Laser Doppler Vibrometer (LDV, Polytech VGO-200).

4. Signal processing approach

4.1. Variational mode decomposition for denoising multi-sensor signals

Noise is an inevitable component that gets entrained into the signals recorded by the sensors used to capture and analyze the dynamics in the machining environment. Removal of noise is a challenging process in micromilling, as it may lead to loss of low amplitude chatter data, if proper cutoff frequency is not selected. Isolating and selecting the

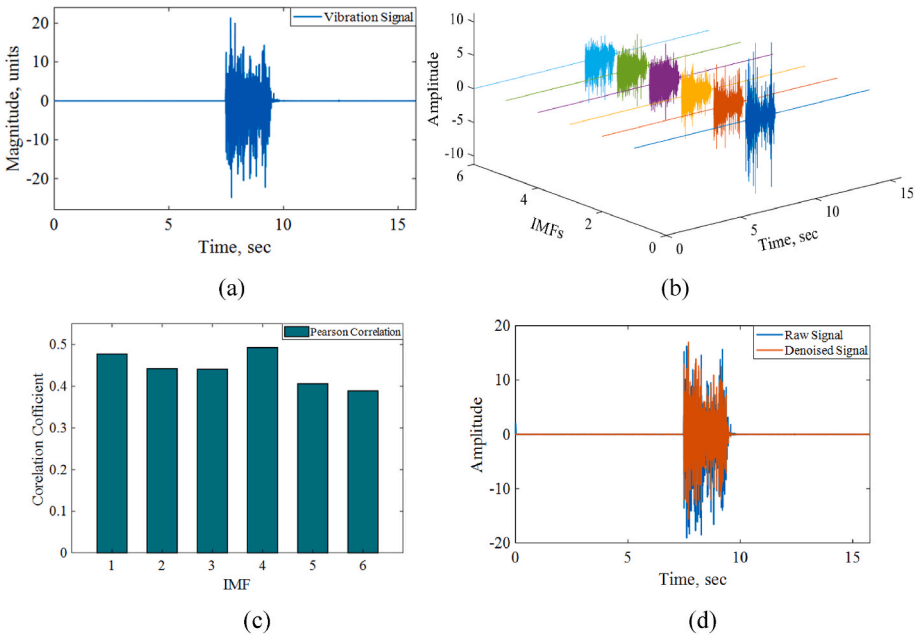


Fig. 3. VMD denoising scheme for the cutting condition E3 (see Table 1); (a) Raw vibration signal; (b) IMFs of the vibration signal; (c) Pearson correlation value of IMFs; (d) Denoised signal superposed with raw signal.

necessary frequency components to reconstruct the signal, is an effective way of denoising the signals without losing the chatter data. VMD is a systematic method for breaking down the complex, non-stationary signals into narrow-bands, known as intrinsic mode functions (IMFs) associated to a specific center frequency without mode mixing. This separation results in a higher frequency resolution, that is essential for chatter onset identification (Zhang et al., 2022). The raw vibration and audio signals (see Fig. 3(a)) are decomposed into a series of narrow-band IMFs, as shown in Fig. 3(b). For decomposing the signal into amplitude-frequency modulated components, the sum of the bandwidths of the decomposed modes should be minimized (Dragomiretskiy and Zosso, 2014). Thus, for a constrained variational optimization problem, the minimization objective can be expressed as:

$$min_{u_k,\omega_k} \sum_{k=1}^{K} \left\| \partial_t \left(\delta(t) + \frac{j}{\pi t} \right) * \left(u_k(t) e^{-j\omega_k t} \right) \right\|_2^2 \quad (1)$$

$$subjected\ to: \sum_{k}^{K} u_k = f(t) \quad (2)$$

where, $\{u_k\}:]= \{u_1, u_2, ..., u_K\}$ represents the extracted IMFs, K is the total number of IMFs extracted, $\{\omega_k\}:=\{\omega_1, \omega_2, ... , \omega_K\}$ represents the center frequencies of each mode, $\delta(t)$ is the Dirac delta function, $*$ represents the convolution, $f(t)$ is the raw input signal in time domain. This objective function minimizes the total bandwidth of the modes by calculating the gradient (partial derivative) of the analytic signal corresponding to each mode. Kurtosis is used to evaluate the IMFs and determine the optimal number of modes. Note that kurtosis is a parameter that measures the distribution or spread of the data and its deviation from the normal distribution. In the present work, kurtosis is used to evaluate the non-Gaussianity of the decomposed components. A higher kurtosis value suggests that the mode captures more informative and transient features, which are crucial for chatter detection. For each mode, $u_k(t)$ extracted from the signal $f(t)$, kurtosis, Ku is computed to find the optimal number of IMFs required for reconstructing the signal. The kurtosis values of the IMFs are compared, and the number of IMFs with the maximum kurtosis is chosen as the optimal number, K. The kurtosis, Ku is calculated as:

$$Ku(u_k) = \frac{1}{K} \sum_{t=1}^{K} \left(\frac{u_k(t) - \overline{u_k}}{\sigma_k} \right)^4 \quad (3)$$

where, K is the total number of extracted IMFs, σ_k is the standard deviation of the $u_k(t)$, $\overline{u_k}$ is the mean of $u_k(t)$. Low-frequency IMFs capture mechanical vibrations, which are critical for understanding the interaction between the tool and workpiece. These frequencies are directly associated with the cutting dynamics, where chatter often manifests as a shift in frequency or amplitude. High-frequency IMFs, on the other hand, capture acoustic emissions, which are essential for cutting tool state monitoring (Kvinevskiy et al., 2020). After selecting the optimal number of modes K based on kurtosis, Ku; the Pearson correlation, P_{corr} between the original signal and each IMF is computed. Pearson correlation coefficient measures the linear relationship between each IMF and the raw input signal. The value of the correlation coefficient helps in identifying the desired frequency components for reconstruction of the signal. Modes with high correlation values are retained for reconstruction, while noise-dominated modes with low correlation are discarded. Mathematically, for the raw input signal, $f(t)$ and the kth IMF, $u_k(t)$; the Pearson correlation is given by:

$$P_{corr} = \frac{\sum_{k=1}^{K} (f(t)_k - \overline{f(t)})(u_k - \overline{u}_k)}{\sqrt{\sum_{k=1}^{K} (f(t)_k - \overline{f(t)})^2} \times \sqrt{\sum_{k=1}^{K} (u_k - \overline{u}_k)^2}} \quad (4)$$

For efficient removal of noise, a threshold for the correlation is

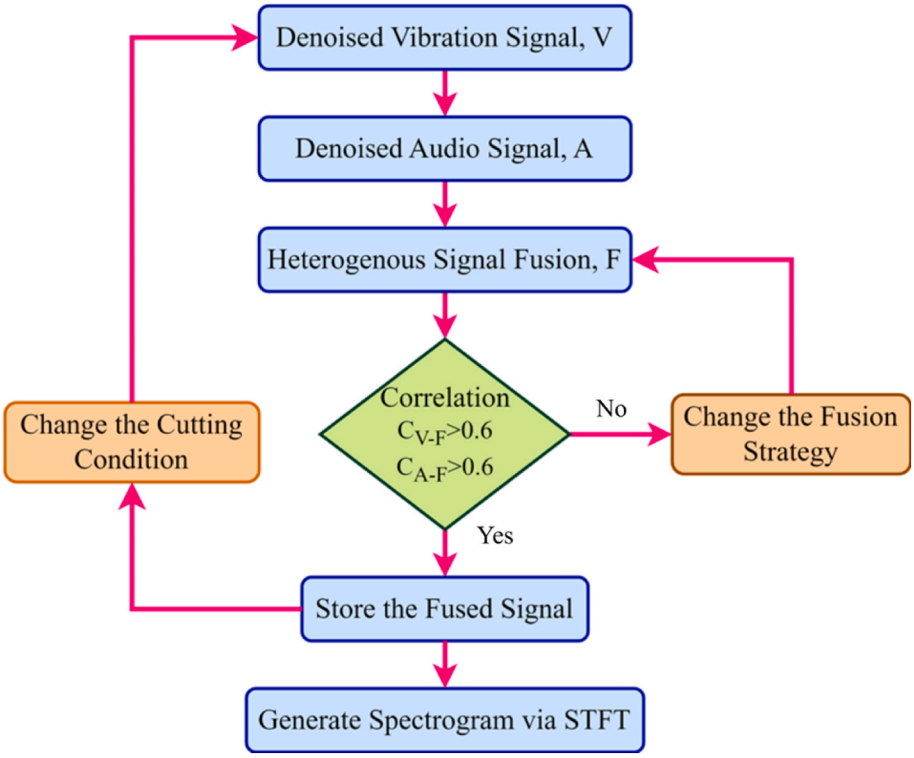


Fig. 4. Algorithm of adopted approach for the fusion of heterogenous signals.

considered. The threshold P_{thresh} is calculated as the root-mean-square (RMS) deviation of the correlation coefficients, meaning it captures the typical deviation from the mean. Only IMFs with correlation coefficients greater than the threshold are considered to reconstruct the signal. The threshold P_{thresh} is give as:

$$P_{thresh} = \sqrt{\frac{1}{K}\sum_{k=1}^{K}(P_k - \overline{P})^2} \tag{5}$$

$$select\ the\ IMF, u_k(t)\ if\ \ P_k\ \ > P_{thresh} \tag{6}$$

where P_k is the computed correlation at kth iteration, and $\overline{P}$ is the mean correlation. The computed correlation is shown in Fig. 3(c). This leads to a set of IMFs, S containing indices of the selected IMFs as:

$$S = \{k \in \{1, 2, ..., K\} | P_k\ > P_{thresh}\} \tag{7}$$

The denoised signal $f(t)_{rec}$ is reconstructed by summing the retained IMFs, leading to a cleaner signal (see Fig. 3(d)) that retains only the most significant components (Zhang et al., 2016).

$$f(t)_{rec} = \sum_{k \in S} u_k(t) \tag{8}$$

4.2. Heterogeneous signal fusion

It is shown in the present work that fusion of signals yields a reliable classification of the cutting conditions into stable or chatter in comparison to single sensor approach. The algorithm of the adopted heterogenous fusion of signals is shown in Fig. 4. Various methods like, fusion of vibration signals, vibration signals fused with cutting force signals, fusion of vibration, audio and cutting force signals are reported. All the signal fusion methods can be grouped into three major categories: data-level fusion, feature-level fusion, and decision-level fusion. Data-level fusion, where raw signals from different sensors are combined before feature extraction, offers the advantage of providing rich, detailed information about the machining process. But it can overwhelm processing systems due to the large volume of data generated, making it impractical for real-time monitoring of the low-immersion micromilling process. Feature-level fusion, combines features extracted from individual sensor signals, reduces data volume and simplifies processing but may fail to capture and present the low amplitude chatter data as feature reduction methods are applied to select few representative features. Finally, decision-level fusion combines the results of separate classifiers, enhancing system robustness but at the expense of increased computational time, which reduces its application in real-time monitoring of high-speed machining. Note that all the reported strategies are offline methods, where the analysis is done post machining making it unreliable for real-time monitoring. The present work has been carried out in a computational device with an intel i5-9500 processor at 3 GHz paired with 32 GB of RAM encompassing a win-11 64-bit operating system.

The present work overcomes these drawbacks by proposing an efficient data level fusion, where the vibration signal, V from an accelerometer (see Fig. 5(a)), and the audio signal, A from the microphone (see Fig. 5(b)) are fused, F (see Fig. 5(c)) using different methods of signal fusion like mean, median, RMS, correlation, and weighted methods. Both the feed direction, V_x and normal-to-feed direction, V_y vibration signals are considered for fusion with audio signals. The quality of the fused signals is estimated by computing the Pearson correlation between the fused signal and individual sensor signal. A threshold of 60 % is fixed for the correlation analysis to select the optimal fusion strategies for deep learning (Fig. 4). Any value of correlation less than 50 % yielded the dominance of individual sensor signal over the fused signal. After the selection of fusion strategies with correlation beyond the threshold, 2D spectrograms of the fused signal is generated using STFTs as shown Fig. 5(d). The spectrogram effectively captures the frequency components of the fused signal over time, which is essential for detecting the transition from stable to chatter condition.

The generated spectrograms are used to train, test and validate the

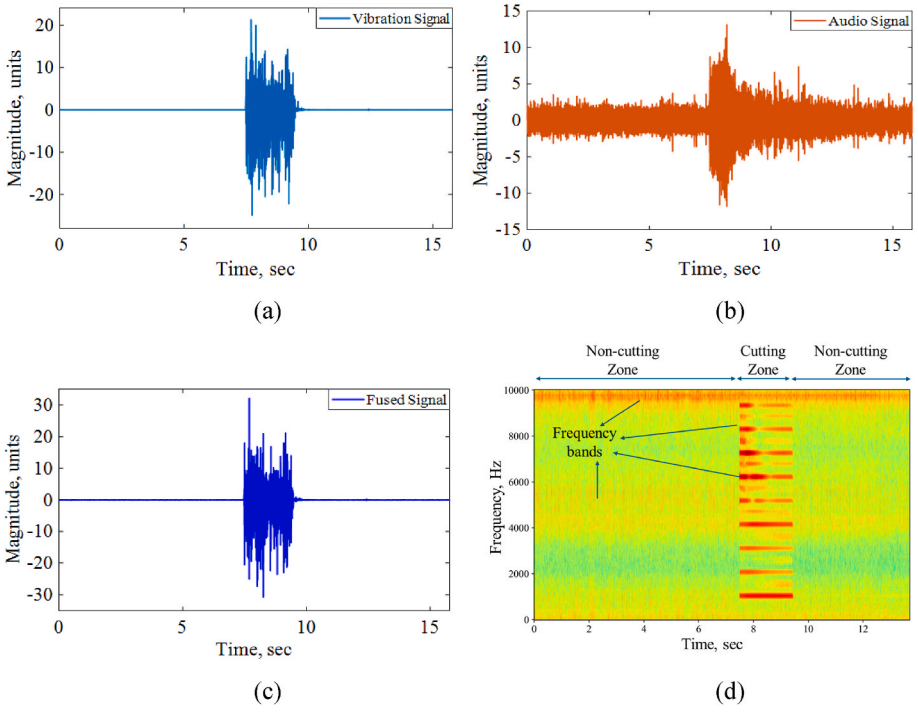


Fig. 5. Time domain fusion of; (a) Vibration signal and (b) Audio signal; to get (c) Fused signal using weighted average strategy; and (d) Spectrogram of the fused signal; for the cutting condition E3 (see Table 1).

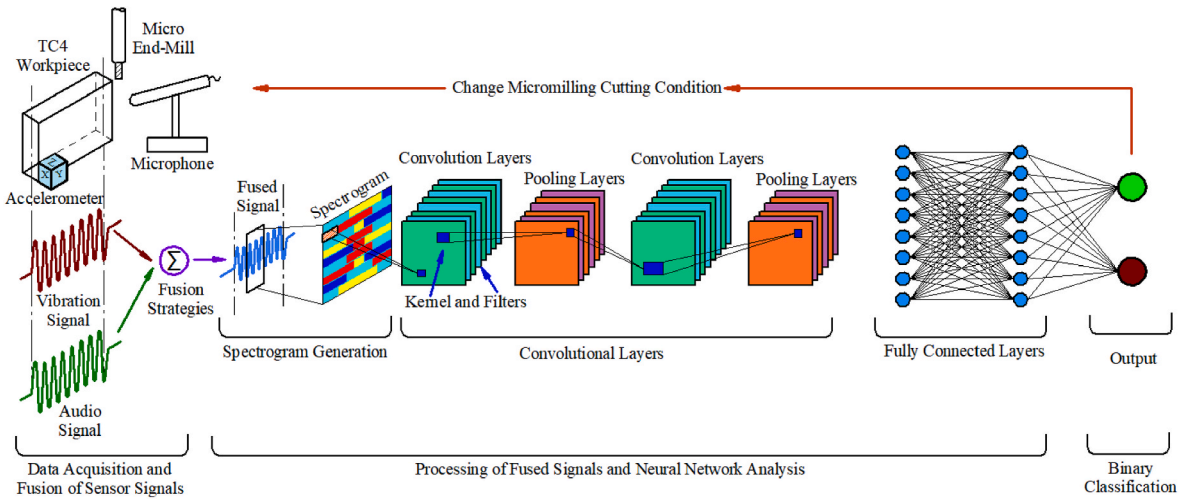


Fig. 6. Schematic of the proposed deep learning strategy for chatter identification.

Table 2
Mathematical expression for various heterogenous sensor fusion strategies used in present work.

Sl No	Fusion Strategy		Mathematical Expression
1	Mean		$F = \frac{V_x + V_y + A}{3}$
2	Median		$F = median\ (V_x, V_y, A)$
3	Root Mean Square (RMS)		$F = \sqrt{\frac{V_x^2 + V_y^2 + A^2}{3}}$
4	Weighted Average		$F = W_1 V_x + W_2 V_y + W_3 A \quad such\ that, W_1 + W_2 + W_3 = 1$
5	Pearson and Spearman Correlation	Median	$F = median\ (V_x, V_y, A)\ subjected\ to \begin{cases} P_{corr}\ between\ (V_x\ and\ A)\ or\ (V_y\ and\ A)\ is\ > T_1\ \ or \\ P_{corr}\ between\ (V_x\ and\ A)\ or\ (V_y\ and\ A)\ is\ > T_2 \end{cases}$
6		RMS	$F = \sqrt{\frac{V_x^2 + V_y^2 + A^2}{3}}\ subjected\ to \begin{cases} P_{corr}\ between\ (V_x\ and\ A)\ or\ (V_y\ and\ A)\ is\ > T_1\ \ or \\ P_{corr}\ between\ (V_x\ and\ A)\ or\ (V_y\ and\ A)\ is\ > T_2 \end{cases}$
7		Weighted Average	$F = W_1 V_x + W_2 V_y + W_3 A \quad such\ that, W_1 + W_2 + W_3 = 1$ $subjected\ to \begin{cases} P_{corr}\ between\ (V_x\ and\ A)\ or\ (V_y\ and\ A)\ is\ > T_1\ \ or \\ P_{corr}\ between\ (V_x\ and\ A)\ or\ (V_y\ and\ A)\ is\ > T_2 \end{cases}$

DL model that has the potential as real-time feedback to monitor the cutting process. The schematic of the proposed deep learning strategy for chatter identification is shown in Fig. 6. In the present work, spectrograms from individual sensor are used for training and the fused signal is used for testing the trained model. For training the proposed DL model, each signal is segmented using a window length of 512 and each image is subjected to data augmentation to enhance the predicting capability of the DL model. Thus, for training a total of three signals from accelerometer (feed and normal-to-feed vibration) and microphone (audio) for thirty-two cutting conditions, with 1833 segments for each signal, and five levels of data augmentation results in 8,79,840 spectrograms. Further, a loop of 10 cycles with all kernel regularizations have been considered to tune the hyperparameters of the DL model. Finally, the model layers, and hyperparameters are selected for training, testing and validation. The DL model has several convolutional layers and pooling layers stacked together where kernels or filters extract important features associated with the chatter onset. The extracted spectrograms are then passed through the convolutional layers, where a set of learnable kernels or filters is applied. These convolutional layers extract features from the input spectrogram, allowing the model to identify the local patterns in the data (LeCun et al., 2015). Pooling layers follow each convolutional layer, reducing the spatial dimensions of the data and ensuring that the model remains computationally efficient while retaining the most salient features. After the convolutional and pooling operations, the extracted features are flattened and fed into fully connected layers. These layers perform the final feature integration and classification task. The fully connected layers capture the high-level dependencies between the extracted features, which are essential for distinguishing between stable and chatter conditions. The output layer of the network employs binary classification to predict the stability of the radial micromilling of thin-walled structures. For testing the trained and validated DL model, spectrograms from the fused signals are used. For sixteen different fusion strategies and thirty-two different cutting conditions, a total of 512 spectrograms are generated and tested for stability using the pre-trained and validated model. The purpose of signal segmentation and data augmentation during the training and validation phase is to increase the size of dataset and to balance the classes for unbiassed training, which is necessary to obtain unbiased classification from the DL model. Table 2 provides the mathematical expressions of different fusion methods used in the present work. In Tables 2, V_x is the vibration signal in the feed direction. V_y is the vibration signal in normal-to-feed direction, A is the audio signal, F is the fused signal. W_1, W_2, W_3 for V_x, V_y, and A, respectively are the weights adopted for fusion strategy. P_{corr} is the Pearson correlation coefficient and S_{corr} is the Spearman correlation coefficient. T (T_1-0.5, T_2-0.6) is the threshold for the correlation coefficient. Similar equations as that of

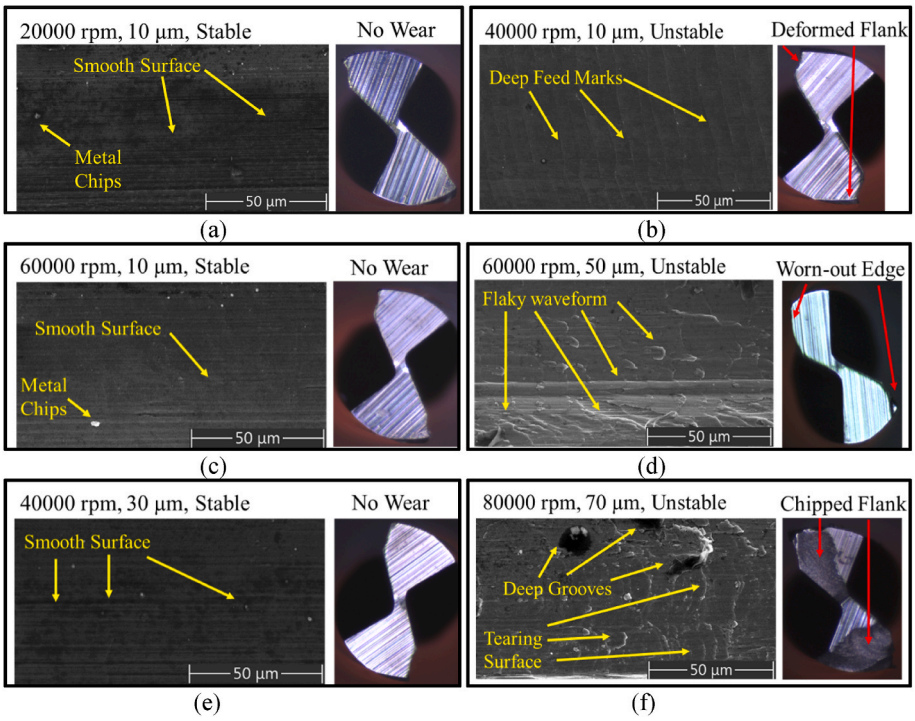


Fig. 7. Machined surface micrograph analysis for labelling the sensor data for training.

Spearman correlation coefficient are used for Pearson correlation coefficient.

5. Results and discussion

5.1. Labelled data training of the DL model

For training the DL model, labelled spectrograms generated from the individual sensor (V_x, V_y, and A) signals are used. Labelling of the spectrogram is done by analyzing the machined surface using SEM micrographs as shown in Fig. 7. The machined surface micrographs are analyzed for formation of any unwanted surface undulations caused due to chatter onset. The deflection of the thin-walled structure increases for every pass of the micro-endmill due to reducing of stiffness. This increased deflection amplifies the vibration thereby imprinting a high wavelength and low-frequency patterns on the machined surface. The presence of these low frequency wavy patterns indicates unstable cutting conditions and the smooth surface without any undulations indicate stable cutting condition (Singh et al., 2015; Gururaja and Singh, 2023). The formation of the pattern depends on the combination of chosen cutting conditions.

The micrograph shown in Fig. 7(a) shows a smooth undisturbed surface, indicating the cutting condition to be stable. Also, the flank on the micro-endmill shows two undamaged cutting edges, complimenting the micrograph. Similarly, for Fig. 7(c) and (e) the micro-endmill passes over the surface smoothly without disturbing the surface. Note that the micro-endmill flank also shows the presence of two-active undeformed cutting edges; hence, the cutting conditions are stable in nature. The low depth of cut provides sufficient damping to the cutting process where the deflections are reduced. However, in Fig. 7(a) and (c) it can be seen that some minute and small metal chips are welded onto the surface due to the increased friction at the cutting zone. Fig. 7(b), (d), and 7(f) shows abrupt changes in the machined surface, where deep feed marks (see Fig. 7(b)) are formed due to the deformed flank surface (see Fig. 7(b)) of the micro-endmill. The flaky waveforms (see Fig. 7(d)) are formed as the cutting edges of the micro-endmill are worn-out (see Fig. 7(d)), due to increased friction and strain hardening. Also, in Fig. 7(f) deep grooves and tearing surfaces are observed as the cutting edge of the micro-endmill gets chipped off (see Fig. 7(f)) due to increased strain hardening attributed to low thermal diffusivity of Ti-alloy at higher spindle speed of 80000 rpm and higher depth of cut 70 µm. The chipped cutting edges of the tool engage abruptly into workpiece surface ploughing it deeply to form groove pockets as shown in Fig. 7(f). The low thermal diffusivity of the titanium alloy increases the temperature and contact stress at the tool–chip interface (primary shear-zone) causing higher friction between the strain-hardened workpiece and the thermally softened tool promoting adhesive wear mechanisms. Furthermore, the strong chemical affinity between Ti-6Al-4V and common tool materials facilitated diffusion wear and chip welding at elevated temperatures (Ezugwu and Wang, 1997; Oliaei and Karpat, 2016, 2017; Zhang et al., 2009). Under these conditions, micro-junctions formed between the workpiece and tool, leading to cold welding of alloy fragments and the development of a thin built-up layer on the cutting edge. Fragments of this layer subsequently fractured and re-adhered to the machined surface, producing the minute welded chips observed in Fig. 7(a) and (c).

Similarly, all the thirty-two conditions are analyzed and classified into fifteen stable conditions and seventeen unstable conditions. These labelled conditions are used to train the DL model. All the spectrogram images are scaled to a uniform size of 224 × 224 x 3, which enables the DL model to easily learn and adapt to the pattern embedded in the fused signals. This square form of images enhances the stability and convergence speed during the training and improves the generalization and prediction accuracy of the DL model. In the present work, rotation and flipping method for data augmentation is used. Since, the signals are already denoised, and the spectrograms are coloured (RGB file), other augmentation methods like scaling, lighting condition, adding salt and pepper noise are ineffective in improving the quality of prediction. Rotation at three different angles of 90°, 180° and 270° are considered with both horizontal and vertical flipping. The adopted training, validation and testing scheme is shown in Fig. 8. In the training phase, the augmented spectrogram images and its corresponding labels, which represent the machining state, are split into training and validation sets using a 70-30 split. This ensures that the model is trained on a sufficient portion of the data while being validated on unseen data to assess its

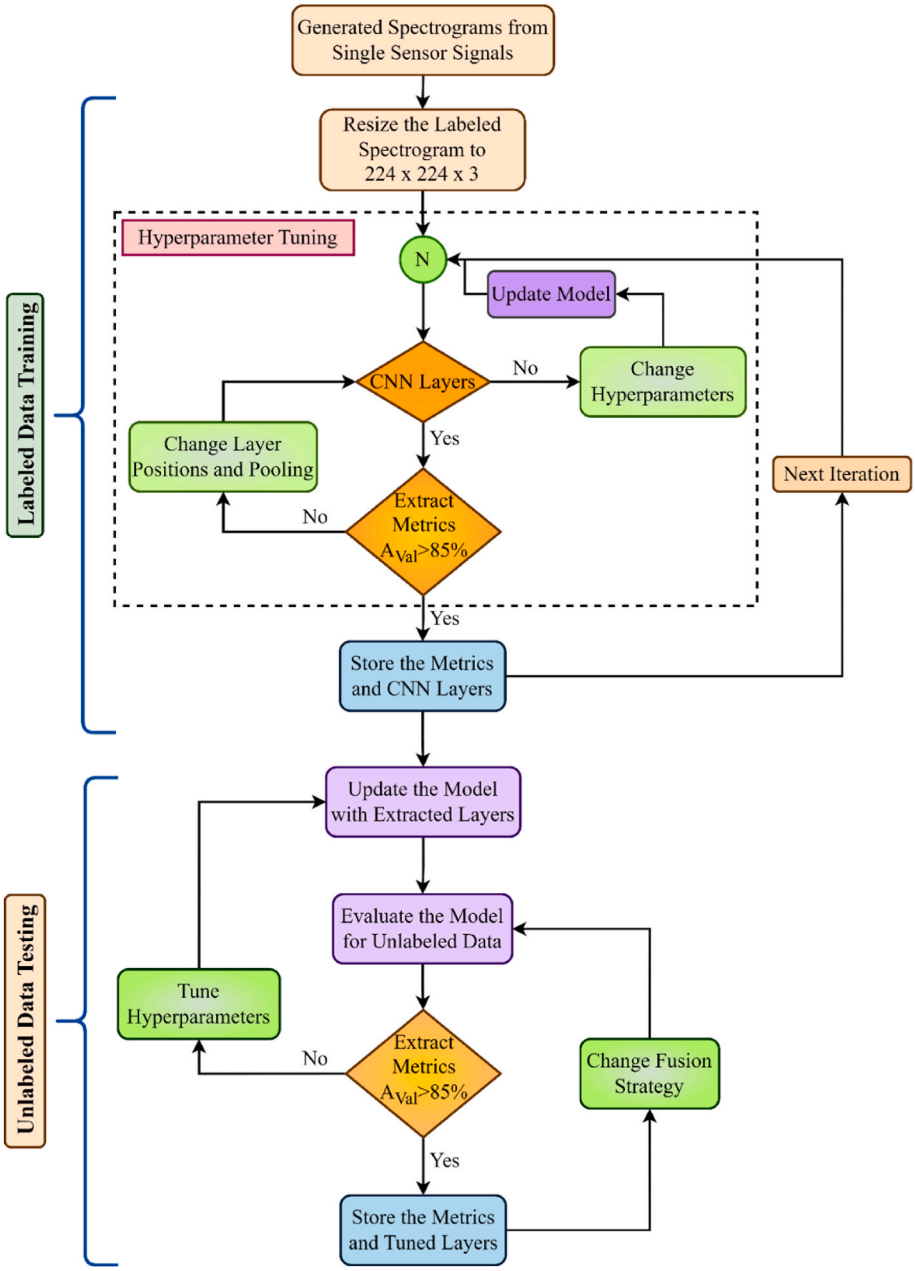


Fig. 8. Algorithm for training, validation, and testing of the DL model.

generalization. The images are then passed through a resizer to ensure the image is in square shape of 224 × 224 x 3. The first convolution layer applies 64 filters of size 3 × 3 with rectified linear unit (ReLU) activation function along with the kernel regularization to prevent overfitting by penalizing large weights.

Next, a max-pooling layer reduces the spatial dimensions of the feature maps by half, retaining the most important features (like high-energy bands in the spectrogram) while reducing the computation load. The feature maps for the cutting condition, E7 (see Fig. 9(a)) and E5 (see Fig. 9(b)) shows the presence of concentrated energy bands at specific frequencies. From Fig. 9(a), it is evident that the spectrogram shows 2nd harmonic of the rotational frequency, 2000 Hz ((N/60) x 2, see E7 in Table 1 for N) as the dominant frequency.

Also note that the spectrograms show the presence of high-frequency bands, but the feature maps do not consider it as a significant feature as seen in the inset of Fig. 9(a). This is because the contact-based accelerometer captures the jig and fixture deflection in addition to the cutting tool-work deflection while machining. Also, the dominant band at 10 kHz represents the sudden change in the frequency of the captured audio while the micro-endmill enters and exits the tool. Also, the insignificant band at 0 Hz is the disturbance induced by flying chips, that pass in the direction of the microphone's diaphragm. Similarly, in addition to the bands of rotational harmonics, a high energy band at 9th harmonic of rotational frequency, 6000 Hz ((N/60) x 9, see E5 in Table 1 for N) is observed in Fig. 9(b). Along with the dominant band a similar high-frequency band at 10 kHz and low-frequency band at 0 Hz is observed as similar to previous instance.

This hierarchical feature extraction process in the pooling layer is followed by a second convolutional layer with 128 filters, and a second pooling layer is applied to further reduce the dimensionality. The arrangement of these layers can be altered to suite the application. In the present study, two convolutional layers with two pooling layers are used. After the convolutional and pooling layers, the feature maps are flattened into a 1D vector. This vector serves as input to the fully

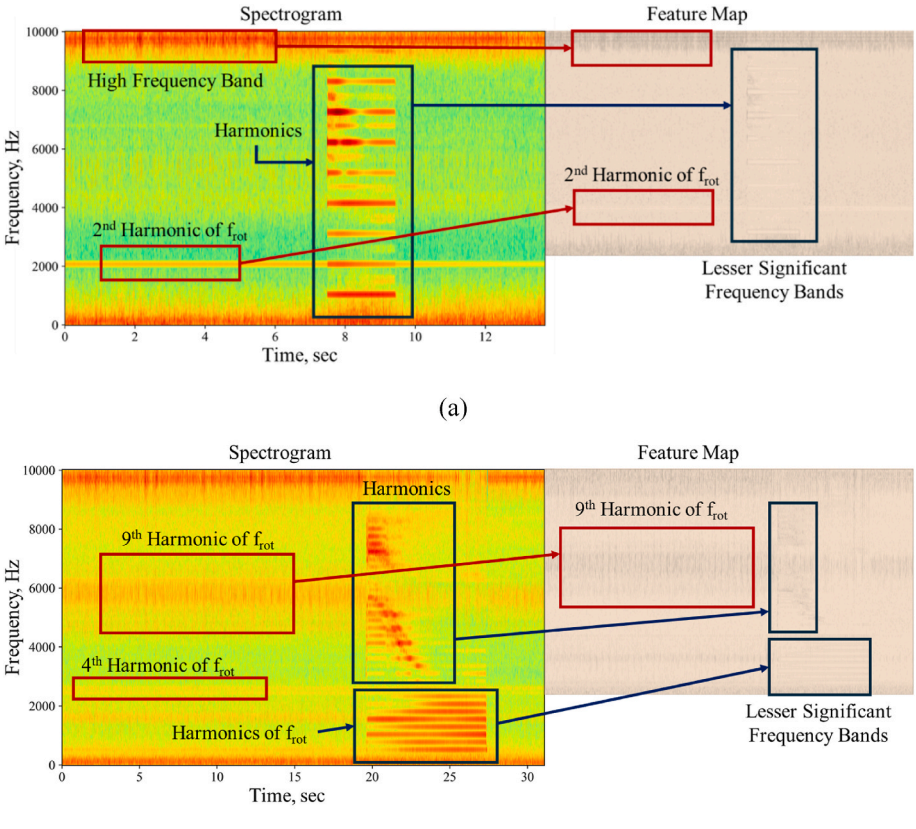


Fig. 9. Feature maps of the fused signal considered by the CNN model for feature extraction for the cutting condition: (a) Stable, E7 (see Table 1); (b) Unstable, E5 (see Table 1).

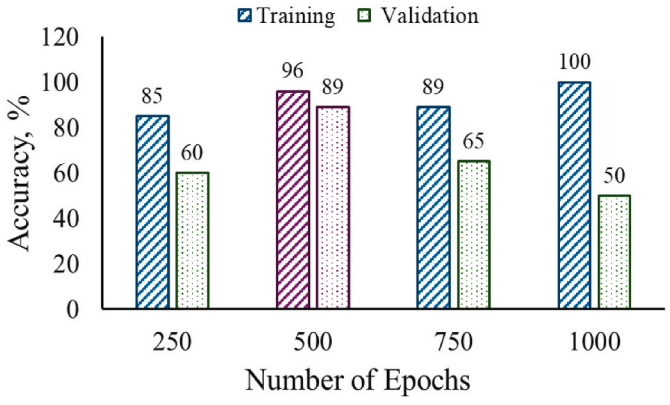


Fig. 10. Variation of accuracy as a function of epoch.

connected layers, where each node connects to all previous activations. The fully connected layer consists of 256 neurons with ReLU activation function along with the kernel regularization, followed by a dropout layer with a 50 % dropout rate to prevent overfitting. The final dense layer has an output size equal to the number of classes, with softmax activation to output class probabilities for stable and chatter conditions. The model is compiled using the Adam optimizer, which adapts the learning rate during training. The loss function is categorical cross-entropy. A learning rate scheduler is defined to reduce the learning rate by 5 % after each epoch. This approach helps the model converge smoothly by making smaller updates to the weights as training progresses. The entire layered model runs for 5 times to optimize all the hyperparameters: Pooling size, Kernel size, Number of filters, Number of epochs, and Batch size. For each run, four levels of epochs are used, 50, 100, 150, and 200. Effectively after 5 runs, the number of epochs will be 250, 500, 750, and 1000. To avoid overfitting while training of the model, Early stopping monitors are used. The model checkpoint callback saves the best model based on validation loss, ensuring that only the most optimal model is retained. Additionally, the learning rate scheduler ensures that training becomes more fine-tuned as the model learns. The best model is chosen based on the highest validation accuracy among the trained models.

The comparison of training and validation accuracy across 250, 500, 750, and 1000 epochs shown in Fig. 10 reveals that 500 epochs is the optimal choice for this model in terms of generalization and avoiding overfitting. At 250 epochs, the model exhibits underfitting, as indicated by the significant gap between training accuracy (85 %) and validation accuracy (60 %), suggesting that the model has not yet fully learned the data patterns. By 500 epochs, the gap narrows significantly, with training accuracy reaching 96 % and validation accuracy at 89 %, which indicates an optimal balance between learning and generalization. Research indicates that when training and validation metrics converge, the model's performance is likely at its best (Ian Goodfellow and Bengio, 2017). Continuing to 750 epochs results in a slight increase in validation accuracy to 65 %, but training accuracy remains high at 89 %, indicating overfitting as the model fails to generalize better to the validation data. At 1000 epochs, training accuracy peaks at 100 %, while validation accuracy drastically drops to 50 %, a clear sign of severe overfitting, where the model has memorized the training data but failed to generalize. This is consistent with findings that suggest the need for early stopping mechanisms to avoid overfitting (Zhao et al., 2020; Huang et al., 2020) Therefore, selecting 500 epochs ensures the best model performance, as further training leads to diminishing returns, and stopping early at this point maximizes generalization to unseen data (Ian

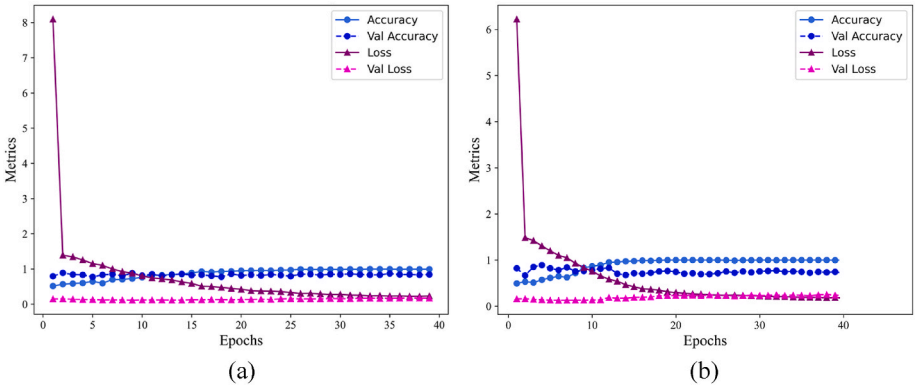


Fig. 11. Loss and accuracy plot of trained CNN model for different fusion strategy; (a) Weighted average; (b) 0.6_PC_Median.

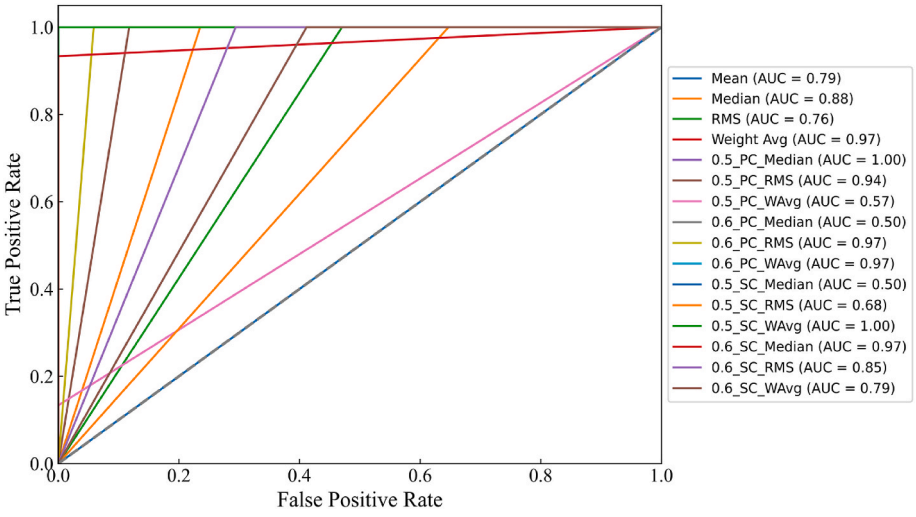


Fig. 12. ROC-AUC curve or different fusion strategies.

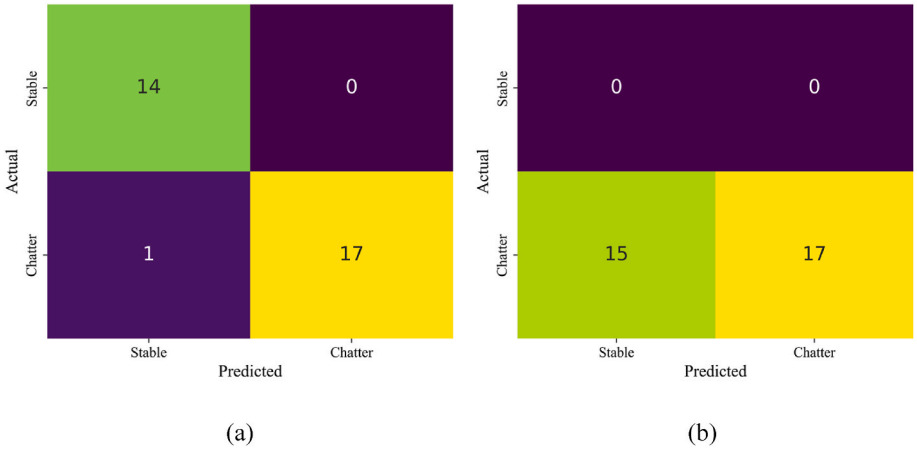


Fig. 13. Confusion matrix of trained CNN model for different fusion strategy; (a) Weighted average; (b) 0.6_PC_Median.

Goodfellow and Bengio, 2017).

5.2. Unlabelled data evaluation of the DL model for instability prediction

The CNN model is tested with the optimized hyperparameters for all the sixteen fusion strategies adopted, and the spectrograms are generated without any signal segmentation. The early stopping is adopted to stop the testing of the model, when the difference between the validation accuracy and accuracy obtained while testing is less than 5 %. Fig. 11 shows the variation of the accuracy and loss for different fusion strategy.

In Fig. 11(a), corresponding to the weighted average strategy, the training and validation loss curves exhibited stable behaviour after the 10th epoch. The validation loss converged at approximately 0.5, while

the training loss stabilized around 0.3. Similarly, the accuracy curves demonstrated consistent improvement, reaching 95 % accuracy for both training and validation datasets around 40 epochs. Conversely, in Fig. 11(b), representing the 0.6 threshold Pearson correlated median-based fusion strategy (0.6_PC_Median), the model displayed slightly better validation performance, with the validation loss approaching 0.4, and the training accuracy stabilizing at 96 %. Despite the better learning performance of the 0.6_PC_Median method across epochs, the model trained with the weighted average fusion strategy exhibited superior generalization capabilities, for unseen data as the difference between the validation and test accuracy is less than 5 %.

The plot shown in Fig. 12 represents the ROC-AUC curves for different fusion strategies used in the CNN model for chatter detection. As indicated, the model performance varied across strategies, with the 0.5_PC_Median and 0.5 threshold Spearman correlated weighted average-based fusion strategy (0.5_SC_WAvg) strategies achieving the highest AUC scores of 1.0, indicating perfect classification capability. Other strategies such as 0.6_PC_RMS and 0.6_PC_WAvg also demonstrated high AUC values of 0.97, showing strong performance in distinguishing between true positives and false positives. Conversely, methods like 0.6_PC_Median and 0.5_PC_WAvg underperformed, with AUC scores of 0.50 and 0.57, respectively, indicating random or weak classification capabilities. The weighted average method also exhibited an AUC score of 0.97, highlighting that the strategy maintains a balance between true positives and false positives. Since, most of the strategies exhibit AUC score >95 %, additional metrics are necessary to select the optimal fusion strategy.

The confusion matrix plot shown in Fig. 13 provides a visual summary of the classification performance of the CNN model when tested with different fusion strategies. In the case of the weighted average fusion strategy, depicted in Fig. 13(a), the model demonstrated a strong classification accuracy by correctly predicting 14 stable cases and 17 chatter cases, with only one misclassified instance of chatter. This performance highlights the model's superior ability to distinguish between stable and chatter conditions, further indicating a high level of generalization in real-world scenarios. The minimal number of misclassifications reflects the robustness of the weighted average fusion method, making it particularly effective in avoiding false positives and false negatives. This balance between detecting chatter and maintaining stability supports the notion that the weighted average strategy is the optimal fusion technique for this dataset (Lamraoui et al., 2015; Tran et al., 2020; LeCun et al., 2015). In contrast, Fig. 13(b) presents the confusion matrix for the 0.6_PC_Median fusion strategy, where the model's performance is considerably weaker in identifying stable conditions. Out of 32 total instances, 15 stable states were incorrectly classified as chatter, resulting in higher false positive rates. While this method did manage to classify all chatter states correctly, the misclassification of stable states suggests that the 0.6_PC_Median strategy may overfit towards detecting chatter at the expense of stability detection. This imbalance makes it less practical for real-time applications where, avoiding unnecessary interruptions due to misclassified stable conditions is critical. Compared to the weighted average strategy, the 0.6_PC_Median approach demonstrates a significant drop in practical utility due to its compromised stability detection (Tran et al., 2020; Zhao et al., 2020; LeCun et al., 2015).

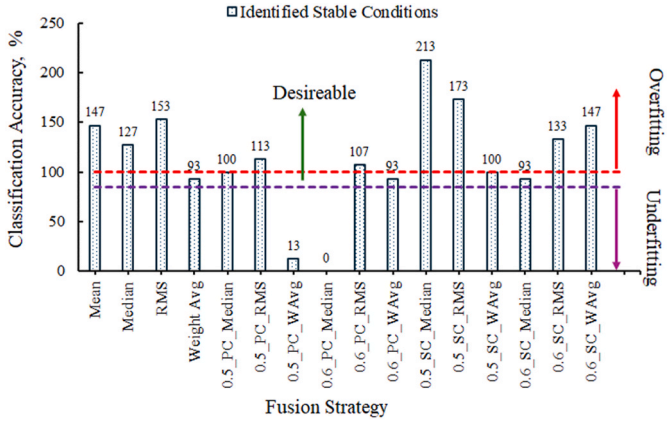


Fig. 14. Classification accuracy of the model with unlabelled data for different fusion strategies.

During the pseudo-labeling phase, the threshold for selecting high-confidence predictions is set at a softmax probability of ≥0.90. This threshold is consistently applied across all operating conditions, including micromilling and turning processes. The selected threshold of ≥0.90 has been informed by empirical analysis conducted on validation subsets, aligning with established and validated practices in deep learning to ensure reliable pseudo-labeling and minimize label noise (Lee, 2013; Sohn et al., 2020). Following the pseudo-label assignment, fine-tuning has been performed on both the labelled and newly pseudo-labelled datasets. To balance contributions from true labels and pseudo-labels, an entropy-minimization loss function has been utilized, adjusting its weights based on prediction confidence. Specifically, the loss weights are dynamically scaled such that pseudo-labelled data points with softmax scores near the threshold (0.90) are assigned lower weights compared to those closer to certainty (softmax ≥0.95), as recommended by contemporary research on adaptive weighting in semi-supervised learning (Sohn et al., 2020; Arazo et al., 2020). Thus, samples exceeding 0.95 confidence are weighted equivalently to manually annotated instances, whereas samples ranging from 0.90 to 0.95 were assigned proportional lower weights. Termination of the fine-tuning stage has been controlled by an early stopping criterion to prevent overfitting and maintain model generalization capability. Training ceased if no improvement in validation loss occurred over ten consecutive epochs, ensuring computational efficiency and stable model convergence (Ian Goodfellow and Bengio, 2017; Géron, 2022). To further enhance reproducibility, a pseudocode detailing each step of the pseudo-labeling strategy, including threshold settings, weight adjustments, and early stopping criteria, has been provided in Appendix B.

Fig. 14 presents the classification accuracy of the CNN model using unlabelled data under different fusion strategies. As shown, certain strategies like 0.5_SC_Median and 0.5_SC_RMS achieved the highest classification accuracy, with values of 213 % and 173 %, respectively. These results suggest that while the 0.5_SC_Median method may excel in identifying specific conditions, it risks overfitting, which can impair its generalizability to unseen data. However, strategies like 0.5_SC_WAvg, 0.5_PC_WAvg and 0.6_PC_Median exhibited significant underfitting, as indicated by their accuracy percentages below 100 %, with 13 % and 0 %, respectively. The weighted average strategy demonstrated a balanced performance, achieving 100 % classification accuracy without any significant overfitting or underfitting issues. The chart demonstrates the effectiveness of fusion strategies in influencing the model's ability to generalize well across unlabelled data, underscoring the importance of selecting an optimal fusion method to avoid overfitting or underfitting, as seen in other fusion strategies.

5.3. Selection of optimal fusion strategy

The evaluation of the CNN model across various fusion strategies reveals a complex picture, making it challenging to definitively identify the optimal fusion strategy due to differing results across the evaluated metrics. Hence, to identify the optimal fusion strategy partial dependence (PDP) and Individual conditional expectation (ICE) plots are generated as shown in Fig. 15.

The given image represents combined Partial Dependence Plot (PDP) and Individual Conditional Expectation (ICE) curves for the weighted average fusion strategy using accuracy (see Fig. 15(a)), recall (see Fig. 15(b)), and precision (see Fig. 15(c)) metrics. The PDP and ICE plots are key tools for understanding the effect of a specific feature on model

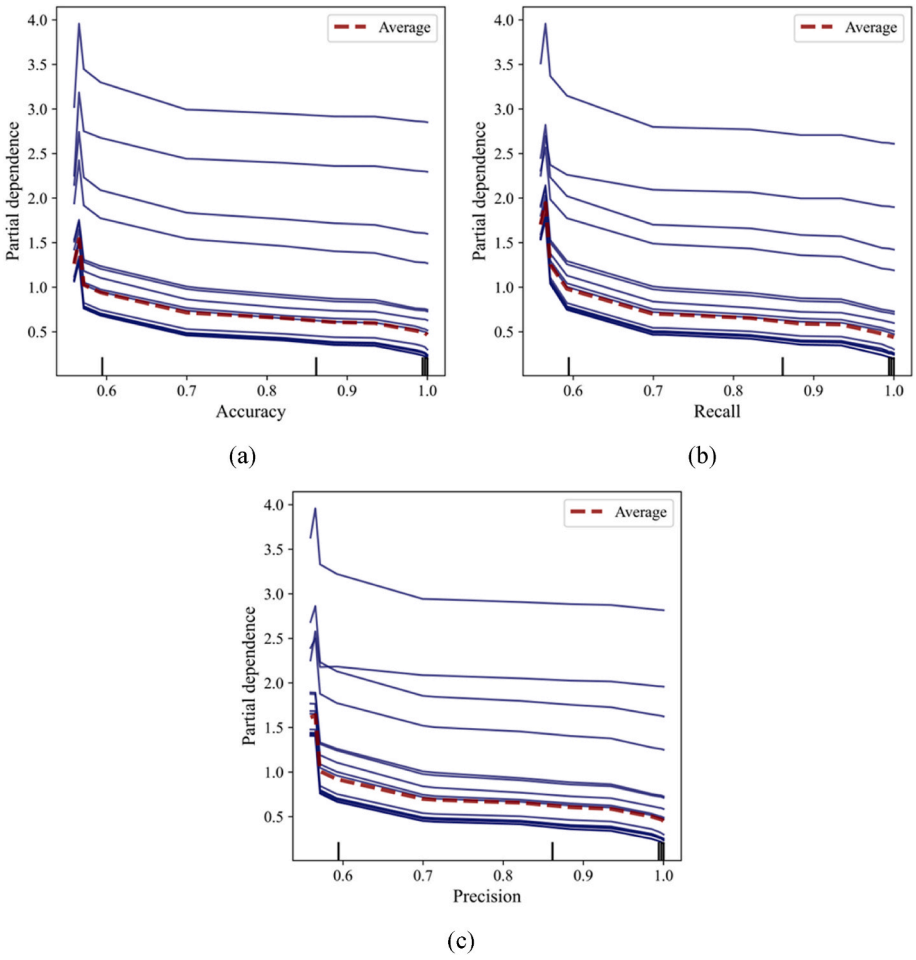


Fig. 15. PDP and ICE plots of for weighted average fusion strategy for different features like; (a) Accuracy; (b) Recall; (c) Precision.

Table 3
Prediction error analysis parameters for ranking different fusion strategy.

Fusion Strategy	MSE	MAE	R^2	RMSE	Explained Variance
Weight Avg	0.001248	0.027416	0.986918	0.035329	0.986918
0.6_SC_WAvg	0.001975	0.037878	0.984623	0.044443	0.984623
0.6_SC_Median	0.002357	0.039954	0.979624	0.048551	0.979624
0.5_SC_Median	0.002878	0.045199	0.96809	0.053645	0.96809
0.5_PC_RMS	0.003976	0.04744	0.960336	0.063058	0.960336
0.6_PC_Median	0.003992	0.051526	0.957503	0.06318	0.957503
Mean	0.00406	0.051916	0.950235	0.063714	0.950235
Median	0.004699	0.05355	0.949392	0.068553	0.949392
0.5_PC_Median	0.008257	0.075821	0.933332	0.090868	0.933332
0.5_SC_WAvg	0.008436	0.083008	0.923803	0.091846	0.923803
0.5_PC_WAvg	0.030715	0.084724	0.687	0.175256	0.687
0.6_PC_WAvg	0.03824	0.098599	0.551221	0.195552	0.551221
RMS	0.118066	0.166261	0.405796	0.343607	0.405796
0.6_PC_RMS	0.122389	0.170614	−0.18031	0.349841	−0.18031
0.6_SC_RMS	0.133254	0.204685	−0.507624	0.36504	−0.507624
0.5_SC_RMS	0.267815	0.218137	−1.984032	0.517508	−1.984032

predictions across the performance metrics. In Fig. 15(a), accuracy is visualized along with the partial dependence values, where the average (represented by the dashed line) gradually decreases from a value near 5 to around 1 as accuracy increases from 0.5 to 1.0. This suggests that higher accuracy is strongly associated with reduced partial dependence, indicating a better generalization of the model as accuracy improves. The variation between individual ICE curves in this plot reflects the diverse impact of different samples on the overall model behaviour, particularly when accuracy exceeds 0.8. The plot shown in Fig. 15(b) shows a similar trend for recall, where partial dependence again decreases steadily as recall improves from 0.5 to 1.0. The dashed line representing the average, which shows a decline from about 5 to nearly 1, indicating that higher recall values result in lower partial dependence, signifying an improved model sensitivity. As with the accuracy plot, the recall ICE curves exhibit variation, but the trend remains consistent, reinforcing the model's stable behaviour across various samples when using the weighted average fusion strategy. This suggests that the model is not only achieving good predictive performance but is also capturing relevant features in the data. In Fig. 15(c), precision is examined. The partial dependence decreases as precision rises from 0.5 to 1.0, with the

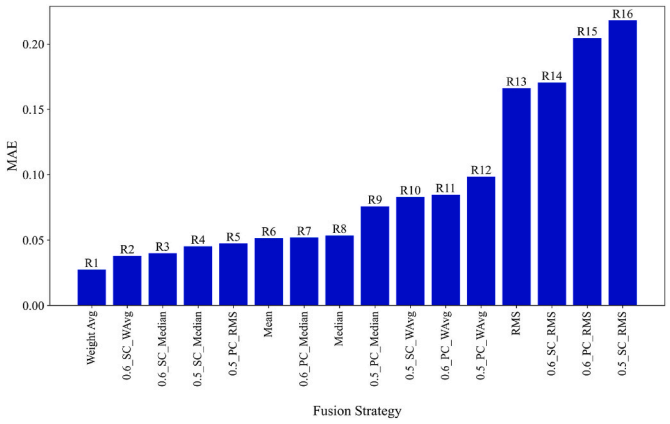


Fig. 16. Ranking of different fusion strategies based on the mean absolute error (MAE).

Table 4
Architecture of CNN model with top ranked fusion strategy.

Layer	Input Shape	Output Shape	Params
Conv 2D	(224, 224, 3)	(222, 222, 64)	1792
Max Pooling 2D	(222, 222, 64)	(111, 111, 64)	0
Conv 2D	(111, 111, 64)	(109, 109, 128)	73,856
Max Pooling 2D	(109, 109, 128)	(54, 54, 128)	0
Flatten	(54, 54, 128)	(373248)	0
Dense	373248	(256)	95,551,744
Dropout	(256)	(256)	0
Dense	(256)	(2)	514

average following a similar trend to accuracy and recall, starting around 5 and stabilizing near 1. This indicates that precision increases correlate with reduced model uncertainty. The spread between ICE curves narrows as precision approaches 1.0, suggesting that the model's behaviour becomes more consistent with higher precision, a desirable characteristic in ensuring fewer false positives in the prediction process. This consistency underscores the effectiveness of the weighted average fusion strategy in refining model performance.

From PDP and ICE plots, it can be noted that the weighted average is the optimal fusion strategy with recall as the best performance metric for evaluation. However, to validate the outcome of PDP and ICE plot, prediction error analysis is done to rank the fusion strategies. The performance of various fusion strategies was evaluated using various error metrics, like Mean Absolute Error (MAE), Mean Squared Error (MSE), R-squared (R^2), Root Mean Squared Error (RMSE), and Explained Variance as shown in Table 3. The ranking of the fusion strategy is demonstrated using the obtained MAE in Fig. 16. The weighted average fusion strategy, exhibited the best performance, as indicated by its lowest MAE of 0.027416, a minimal MSE of 0.001248, and an R^2 value of 0.986918, signifying a high level of accuracy in predicting the target values. The DL model's performance based on the weighted average fusion strategy tops the error analysis as it assigns appropriate weights to individual sensors before fusing. This low level of error means that the fusion strategy preserves the underlying trends of the data without allowing outliers or noise to disproportionately affect the final prediction, leading to a more consistent and accurate output. It is interesting to note that the next few strategies are based on the Spearman correlation coefficient, and these strategies are closer to weighted average as seen by the parameters like explained variance, R^2, MAE, and MSE. The weighted average overcomes these strategies by a finer margin, as it tends to filter out the noise by providing appropriate weights to the sensors and enhances the strength of more relevant features, thus providing reliable predictions over other strategies. A detailed parameter table has been provided to summarize the hyperparameters, execution durations, and processing specifics across all methodological steps as shown below in Table 1 of Appendix C.

For the optimal fusion strategy, weighted average, the complete CNN model architecture is as given in Table 4. Similar arrangement of five convolutional layers with pooling layers was reported in the Alexnet model with 60 million parameters (Krizhevsky et al., 2012). However, the proposed model has shown that for reliable classification only two convolutional layers with pooling layers is sufficient, this reduces the computational load significantly. Note that the present model has shown that after flattening, 95.6 million parameters are available, which is higher than the reported value. This higher value may seem to induce overfitting, but in the present work, data augmentation, kernel regularization and Adam optimizer is adopted which nullifies the risk of overfitting. The unseen advantage of having more parameters, is that the model can easily memorize the training data instead of generalizing the unseen data. Also, for higher-dimensional inputs like spectrograms, the additional parameters allow the model to better learn hierarchical

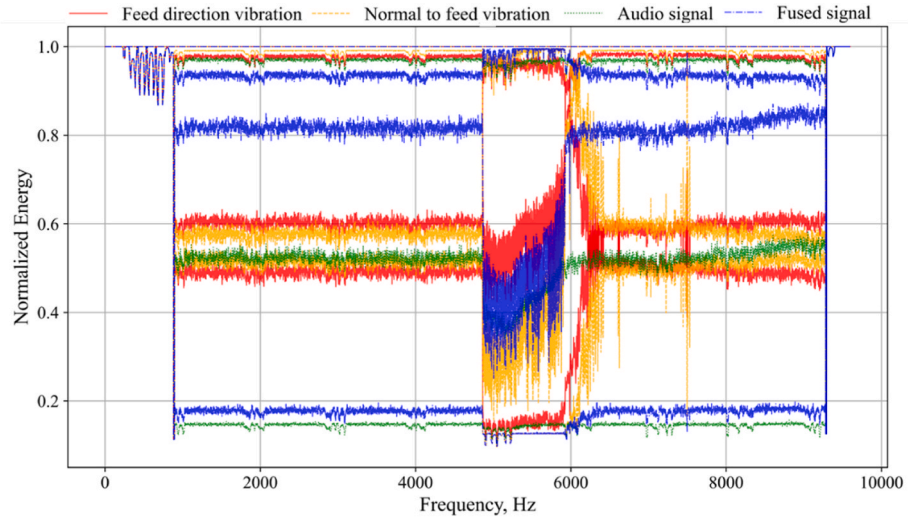


Fig. 17. Normalized energy distribution of signals as a function of frequency.

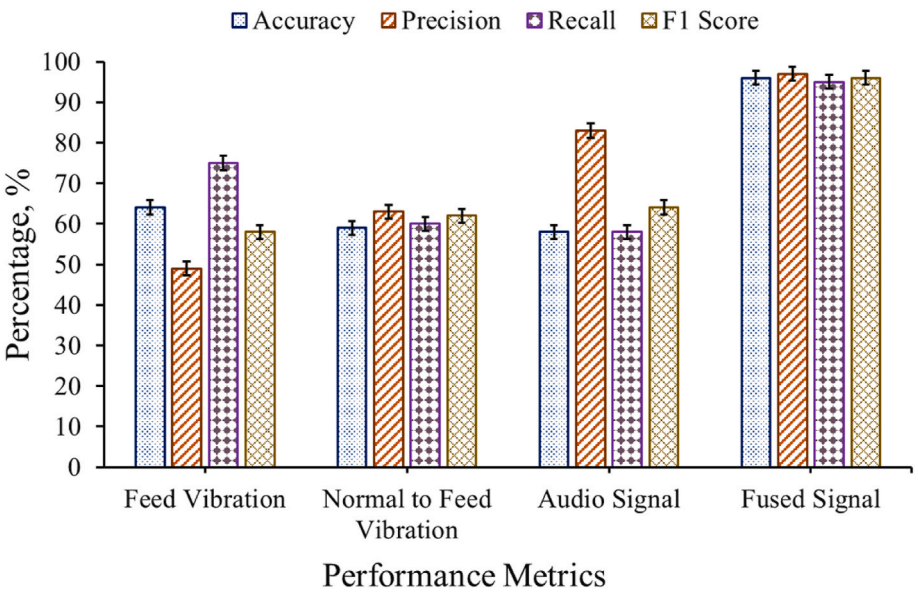


Fig. 18. Performance analysis of the final CNN model for different sensor signals.

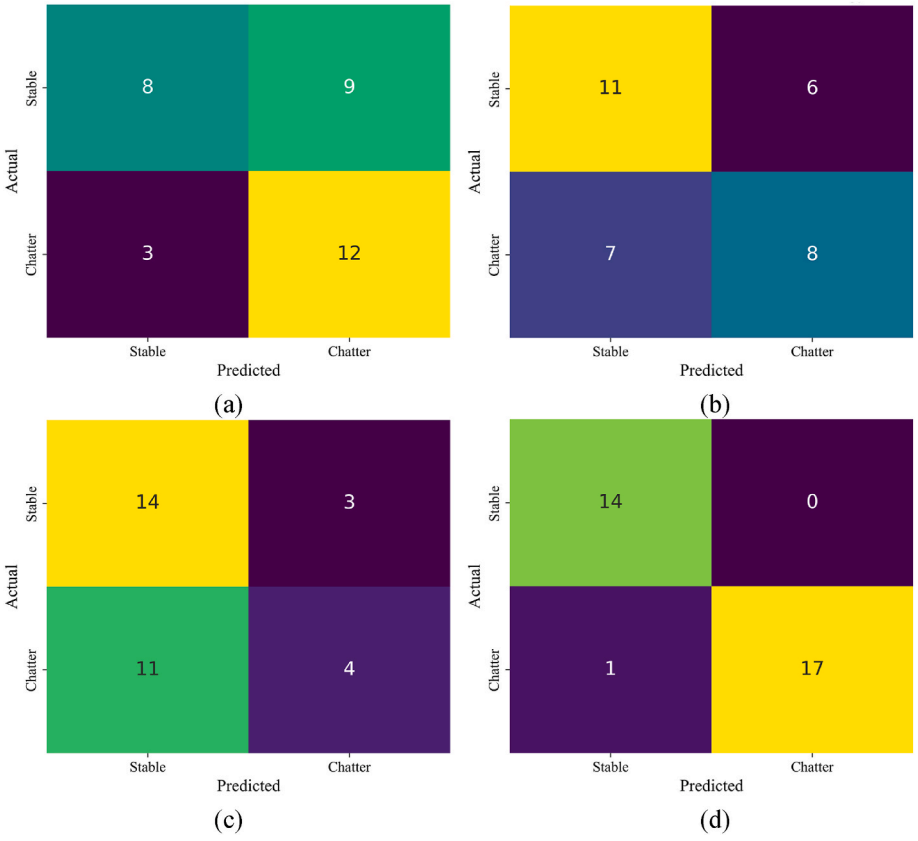


Fig. 19. Confusion matrices of the final CNN model for different sensor signals; (a) Feed vibration; (b) Normal to feed vibration; (c) Audio signal; (d) Fused signal.

feature structures, which can improve the classification accuracy.

5.4. Ablation study of the trained and validated DL model

In the present section, an ablation study is done to show the effectiveness of fusing the vibration and audio signals for chatter onset detection in high-speed micromilling of thin-walled structures. Fig. 17 shows the distribution of the normalized energy extracted and superposed from the single sensor signal and fused signal spectrogram for the cutting condition E3 (see Table 1). Each individual sensor captures specific aspects of the system's behaviour but is limited in its capacity to detect comprehensive energy fluctuations. For instance, the accelerometer captures the dynamics, but it also captures the deflection of the entire tool-work assembly, and the audio sensor effectively captures the cutting dynamics, but does not account for the tool-work deflection induced. This is evident from Fig. 17, as the energy bands

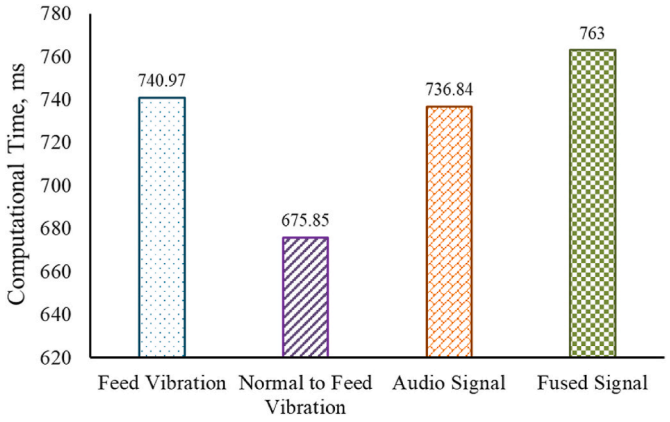


Fig. 20. Computational time analysis of the final CNN model for different sensor signals.

of accelerometer in both the feed and normal-to-feed directions are widely distributed especially in the range between 5000 and 6500 Hz, in comparison to the narrow distribution of the audio signal energy. The highest energy band can be neglected as it generally represents the entry and exit of the cutting tool while machining. The fused signal shown in blue colour captures the machining dynamics more effectively by combining the advantages of both the individual sensors and the weighted average strategy further reduces the noise to retain necessary features of the signals across the entire frequency range.

Table 4 provides the CNN architecture for the optimal fusion strategy, weighted average and from Fig. 17 it is shown that the fusion of signals yields a reliable classification. However, to analyze the effect of fusion on the CNN models performance, a parametric study is also conducted. The outcome is presented in Figs. 18–20, which shows the performance metrics, confusion matrix and the computational time, respectively for the individual sensor signal and fused signal. The fused signal achieves an accuracy of 0.96, precision of 0.97, recall of 0.95, and an F1-score of 0.96, outperforming the single signals across all metrics shown in Fig. 18. In contrast, individual sensor signals such as feed direction vibration, normal-to-feed vibration, and audio signal show significantly lower performance. The improvement in accuracy is 33 % when fused signals are used instead of accelerometers and 39 % improvement when fused signal replaces audio signals. Similar enhancement in precision is 21 % and 39 % and recall shows 29 % and 9 % improvement when fused signal replaces the individual sensor signals. This improvement in fused signal performance can be attributed to its ability to capture a broader range of features across different signal modalities, leading to a more comprehensive representation of the machining dynamics. Further, a 5-fold cross-validation has been conducted on the micromilling dataset, which yielded a mean training and validation accuracy of 0.92 and 0.90, respectively, with a narrow gap (<2 %) across all folds. The standard deviation between the runs is ±1.55 % and the same has been used to plot the error bar as shown in Fig. 18. The confusion matrices for the different signals shown in Fig. 19 highlights the effectiveness of signal fusion as classification accuracy is improved. The feed direction vibration (see Fig. 19(a)) misclassifies 9 stable instances and 3 chatter instances, resulting in a high error rate while normal-to-feed vibration (see Fig. 19(b)) misclassifies 11 stable and 6 chatter instances. The audio signal (see Fig. 19(c)) misclassifies 4 chatter instances. But the fused signal (see Fig. 19(d)) misclassifies only one instance.

For the computational time analysis shown in Fig. 20, it is important to note that the fused signal strategy takes 22 ms more than the feed direction vibration, 87.15 ms more than the normal-to-feed direction vibration, and 26.16 ms more than the audio signal for the classification task. While the proposed fusion strategy results in a slightly longer processing time, the substantial improvement in accuracy, recall, and F1-score outweighs the increased computational cost. The F1 score improved by 32 %, and 28 %, respectively when the fused signal replaces the vibration signal and audio signal.

5.5. Application of the proposed DL framework to other manufacturing processes

5.5.1. Turning of FSS-409 alloy

The trained and validated DL model with optimal fusion strategy has been used to predict the stability of FSS-409 alloy while turning it on HMT-Praga (Model: PTC-200) CNC lathe machine tool at different cutting conditions. Note that since the evaluation of the turning process is done using the spectrograms of the fused signal, no additional hyperparameter tuning or rearranging of the layers is needed. The turning operation has been conducted on FSS-409 at four levels of depth of cut (0.1 mm–0.4 mm at interval of 0.1 mm), four levels of feed rate (0.05 mm/rev to 0.2 mm/rev at interval of 0.05 mm/rev), and at four levels of cutting velocity (50 m/min to 200 m/min at interval of 50 m/min). A total of sixty-four experiments (see Fig. 21) has been carried out and the vibration and audio signals are recorded.

The generated spectrograms are evaluated for the occurrence of chatter and the DL model was able to predict the chatter onset with 95.31 % accuracy, 98 % precision, 97 % recall, and 97 % F1-score. The performance metrics are shown in Fig. 22(a) and the confusion matrix is shown in Fig. 22(b). For the turning experiments, the model exhibited a mean accuracy of 95.31 % with a standard deviation of ±0.94 %, indicating robustness even when exposed to unseen process conditions for a 5-fold cross validation. Table 5 shows the validation of the prediction with the machined surface micrograph. This shows that the proposed framework is capable of handling new unseen data obtained from different machining process as it is able to generalize the prediction using the knowledge gained while training.

In Table 5, CC is cutting condition, D is depth of cut in mm, F is feed rate in mm/rev, and V is cutting velocity in m/min. As the turning process transits from stable cutting to chatter dominated cutting, the

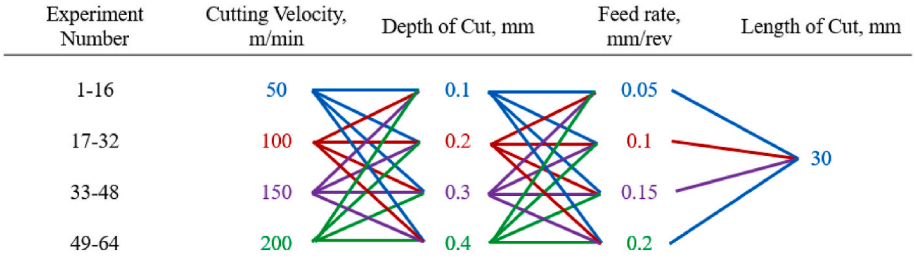


Fig. 21. Ray diagram for sixty-four cutting conditions used in CNC turning of FSS-409 alloy.

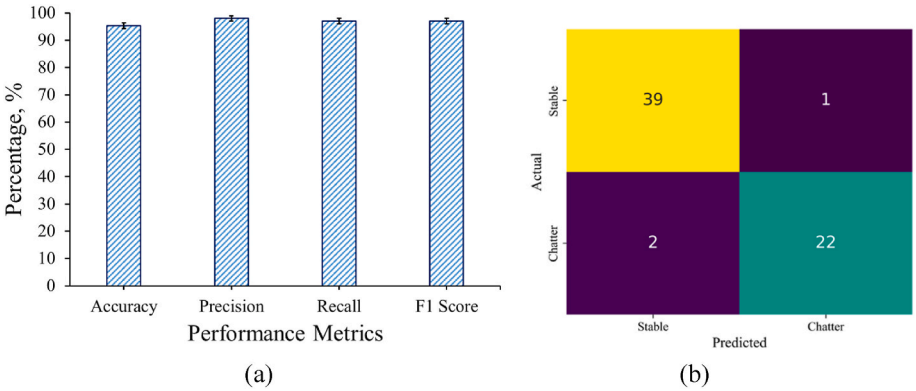
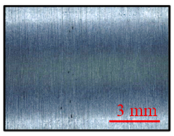
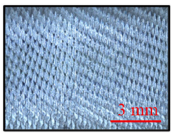
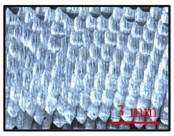
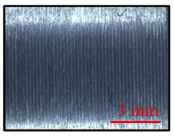
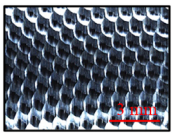
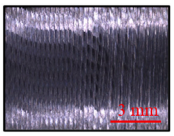
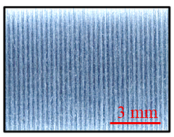
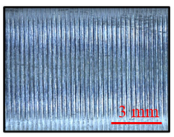


Fig. 22. Extension of the proposed DL framework to FSS-409 turning process with optimum fusion strategy; (a) Performance metrics of the model; (b) Confusion matrix for different cutting conditions.

Table 5
Application of the pretrained DL model to predict stability in turning of FSS-409.

Sl No	Cutting Conditions	Identified Stability from the DL Model	Machined Surface Micrograph	Sl No	Cutting Conditions	Identified Stability from the DL Model	Machined Surface Micrograph
1	CC: E1 D: 0.1 mm F: 0.05, mm/rev V: 50, m/min	Stable		5	CC: E37 D: 0.3 mm F: 0.1, mm/rev V: 50, m/min	Chatter	
2	CC: E10 D: 0.1 mm F: 0.15, mm/rev V: 100, m/min	Chatter		6	CC: E44 D: 0.3 mm F: 0.15, mm/rev V: 200, m/min	Stable	
3	CC: E22 D: 0.2 mm F: 0.1, mm/rev V: 100, m/min	Chatter		7	CC: E52 D: 0.4 mm F: 0.05, mm/rev V: 200, m/min	Stable (Misclassified)	
4	CC: E27 D: 0.2 mm F: 0.15, mm/rev V: 150, m/min	Stable		8	CC: E62 D: 0.4 mm F: 0.2, mm/rev V: 100, m/min	Stable	

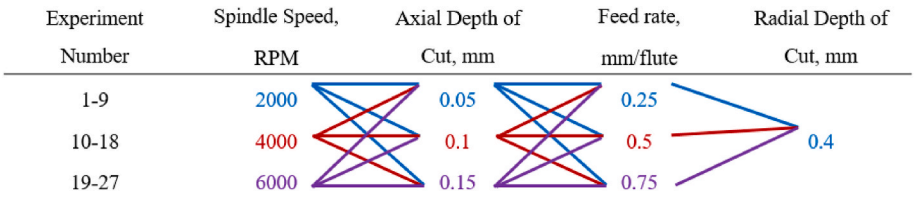


Fig. 23. Ray diagram for twenty-seven cutting conditions used in CNC slot milling of IN-718 alloy.

change in probability of prediction yielded a misclassified test case where chatter dominated cutting is wrongly predicted as stable cutting (see no.7 in Table 5).

5.5.2. *Slotting of IN-718 alloy*

The trained and validated DL model with weighted average fusion strategy (optimal) has been used to predict the stability of the milling process while performing slotting experiments on Inconel-718 alloy on CNC vertical machining center (Model: Hardinge, Bridgeport GX-600) at different cutting conditions. The endmill used is of 4 mm in diameter with 4 flutes. Note that since the evaluation of the slot milling process is done using the spectrograms of the fused signal, no additional hyper-parameter tuning or rearranging of the layers is needed.

The slotting operation has been conducted on IN-718 at three levels of depth of cut (0.05 mm–0.15 mm at interval of 0.05 mm), three levels of feed rate (0.25 mm/flute to 0.75 mm/flute at interval of 0.25 mm/flute), and at three levels of spindle speed (2000 rpm–6000 rpm at interval of 2000 rpm). A total of twenty-seven experiments (see Fig. 23)

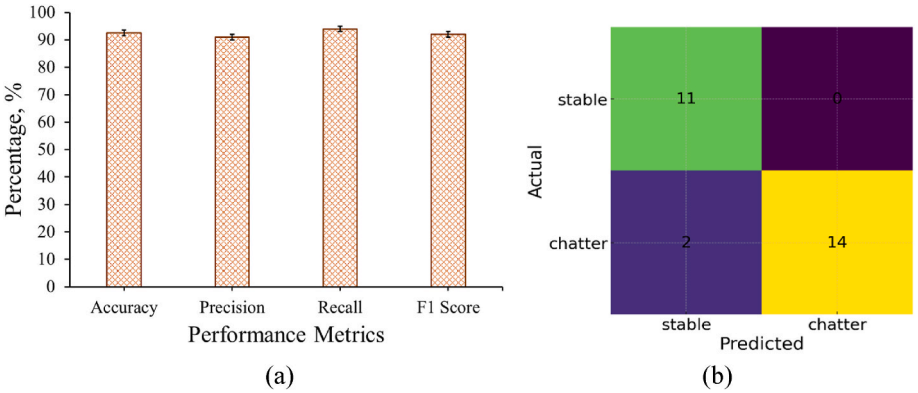
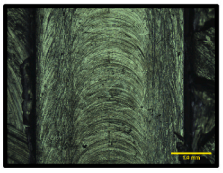
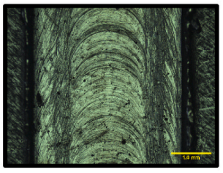
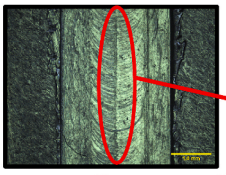
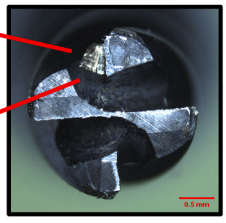


Fig. 24. Extension of the proposed DL framework to IN-718 slot milling process with optimum fusion strategy; (a) Performance metrics of the model; (b) Confusion matrix for different cutting conditions.

Table 6
Application of the pretrained DL model to predict stability in slot milling of IN-718.

Sl No	Cutting Conditions	Identified Stability from the DL Model	Machined Surface Micrograph	Remarks
1	CC: E14 N: 4000 rpm F: 0.5, mm/flute D: 0.1, mm	Stable		
2	CC: E25 N: 6000 rpm F: 0.75, mm/flute D: 0.05, mm	Stable		
3	CC: E2 N: 2000 rpm F: 0.25, mm/flute D: 0.1, mm	Chatter		Wornout Flute 
4	CC: E18 N: 4000 rpm F: 0.75, mm/flute D: 0.15, mm	Stable (Misclassified)		

has been carried out dry condition and the vibration and audio signals are recorded.

The performance metrics shown in the bar plot (see Fig. 24(a)) demonstrates a reliable and consistent classification capability of the developed DL model under varied slot milling conditions. Fig. 24(a) is a validation that the developed model has understood the patterns within the spectrograms and was able to generalize the learned knowledge for the new unseen data, indicating a balanced and robust classification capability. Note that traditional single-sensor techniques, such as RMS-based vibration indicators, have long been used for chatter detection but often required high computational effort or sacrificed generality across cutting conditions (Li et al., 2014). By contrast, the developed DL model achieved an accuracy of 92.6 % with a precision of 91.0 %, recall of 94.0 %, and F1 score of 92.0 % while remaining light enough for real-time chatter identification. Also, a 5-fold cross validation has also been done to understand the model's robustness. For milling conditions, the model yielded a mean accuracy of 92.6 % with a standard deviation of ±0.82 %, and a 95 % confidence interval of [91.78 %, 93.42 %]. Similarly, the recall and precision values demonstrated narrow variability margins, reinforcing the model's consistent performance. Similarly, the confusion-matrix heatmap shown in Fig. 24(b) highlights the DL model's generalizing capability to identify chatter at different cutting conditions. Note that out of 91 test cases across turning (64), and slotting (27), only five misclassifications occurred: three in turning and two in slotting. The reported literatures with decision-level fusion schemes suffers from elevated false-positive rates, leading to false chatter alarms and unwanted process interruptions (Kuntoğlu et al., 2020). The correlation-guided, data-level fusion employed in the present study effectively

suppresses uncorrelated sensor noise while preserving the essential dynamic features from both the sensors (vibration and audio), minimizing the false positive rates. The outcome of the model has been validated with machined surface analysis as shown in Table 6. In Table 6, CC is cutting condition, D is depth of cut in mm, F is feed rate in mm/flute, and N is spindle speed in rpm. The model prediction is in good agreement with the micrograph analysis.

The few residual errors (misclassifications) can be attributed to several factors. First, the transition to chatter from stable condition produces a gradual shift in the dominant frequency component, causing a frequency band overlap that reduces the classifier's confidence (Siddhpura and Paurobally, 2012). Due to this there were three misclassified cases in the turning operation. Second, when uncut chip thickness or depth of cut is low, vibration and acoustic signals approach baseline noise levels, causing low-amplitude chatter signatures to be misinterpreted as noise (Bhuiyan et al., 2016), this explains the two misclassifications in slotting. Third, tool-wear and built-up-edge formations induce erratic high-frequency bursts, that can mimic chatter (Khasawneh et al., 2018). However, in the present study this erratic high-frequency burst effect has been compensated by data augmentation, data segmentation and class-balancing strategies. Despite these occasional misclassifications, the DL model exhibited excellent transferability when deployed without re-training to both FSS-409 turning and IN-718 slotting datasets, achieving over 92 % accuracy with high precision/recall. This consistent and reliable performance of the model indicates that fusing vibration and acoustic signals yields a more generalizable representations of chatter than individual sensor modalities. Consequently, the developed framework overcomes the need for process-specific training or hyperparameter tuning, supporting its adoption as a unified, AI-driven stability-monitoring framework across diverse manufacturing scenarios.

6. Conclusion

The present work proposes a CNN-based interpretable stability estimation framework for high-speed micromilling of thin-walled TC4 alloys. By integrating multi-sensor fusion and deep learning strategies, the study highlights the model's effectiveness in improving the chatter onset detection accuracy making it a reliable solution for real-time monitoring. Major findings from the present study are:

- A DL model has been developed, trained, validated and tested for chatter onset detection in micromilling of thin-walled TC4 alloys with the prediction accuracy of 96.87 %.
- Fusion of vibration and audio signals significantly improves chatter detection. The developed DL model with the optimal fusion strategy yielded a mean absolute error of 0.027416.
- The DL model has been trained with 879,840 spectrograms generated from fused signals, ensuring a robust and generalizable prediction of chatter onset without any bias.
- The DL model has been trained with the labelled data and evaluated on the unlabelled data, which showed 89 % validation accuracy. This shows the proposed models capability to generalize the learning gained in the training to adapt for the unseen data in the testing.
- The weighted average fusion strategy has been identified as the optimal method for multi-sensor signal fusion, outperforming other fusion strategies by 33 % enhancement in the prediction accuracy and achieving an R^2 value of 0.986918.
- Ablation studies confirmed that signal fusion of vibration and audio signals enhanced performance metrics like precision and recall by 21 % and 29 %, respectively compared to individual sensor inputs.
- The developed DL model requires only 500 epochs for optimal training, balancing accuracy and avoiding overfitting.
- The proposed DL model with fused signal showed increased computational time of 22 ms and 26.16 ms, respectively in comparison with the individual sensors: accelerometers and microphone but, a 32 % and 28 % improvement in the F1 score, respectively has been observed when the fused signal replaces the vibration signal and audio signal, respectively.
- The pretrained DL model was able to exactly predict the chatter onset in sixty-one turning operation test cases out of sixty-four test cases as validated by the machined surface micrograph with 95.31 % accuracy.

7. Future scope and limitations

7.1. Future scope

- Building on the current image-based approach, fusing additional sensors—such as acoustic emissions, laser displacement, laser vibration (LDV) and cutting-force measurements—could further strengthen chatter detection across diverse materials and cutting conditions.
- Integrating the proposed algorithm directly into CNC control software as a feedback signal will enhance chatter identification, thereby improving the production efficiency.

7.2. Limitations

- The deep-learning model was trained exclusively on RGB spectrogram images; consequently, any deviation from this input format—such as raw time-series data or alternative visual encodings—would necessitate reconfiguration of the network architecture and re-training to accommodate the new data representation.
- It is essential to ensure that the training set is free from class imbalance to avoid biased output from the model. The skewed data distributions, arising from the class imbalance can lead to biased predictions favouring majority-class even after hyperparameter tuning.

CRediT authorship contribution statement

Sethurao Gururaja: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kundan K. Singh:** Writing – review & editing, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

The pseudocode for the current study is as given below, the analysis of the data is done in python environment (3.12):

```
BEGIN
  Step 1. Initialize Environment
        IMPORT pandas, numpy, matplotlib.pyplot, seaborn, tensorflow.keras, sklearn
        (train_test_split, metrics), cv2, shap
        CONFIGURE global plot settings (font size, DPI)
  Step 2. Load & Preprocess Data
        IF image dataset THEN
          LOAD and RESIZE images → target shape (e.g. 224×224)
          NORMALIZE pixel values (÷255)
          APPLY data augmentation (rotate, zoom, shift, shear, flip)
        ELSE
          LOAD tabular data (CSV/Excel)
          EXTRACT X features and y target
          IMPUTE missing values
          STANDARDIZE features (z-score)
        END
        SAVE preprocessed subsets
  Step 3. Split Data
        SPLIT X, y into X_train, X_test, y_train, y_test (test_size=0.2)
        IF image data THEN SCALE X_train, X_test (÷255)
  Step 4. Build & Compile CNN
        PRINT "Building model..."
        DEFINE CNN:
           Input → Conv2D + ReLU → MaxPool → Dropout → … → Flatten →
        Dense(256, ReLU) → Output(Softmax)
          COMPILE(model, optimizer=Adam(lr=0.001), loss=categorical_crossentropy,
        metrics=[accuracy])
        PRINT "Model ready."
  Step 5. Train Model
        PRINT "Training..."
          SET callbacks: EarlyStopping(patience=3), ModelCheckpoint,
        LearningRateScheduler
          FIT model on training data (epochs=50, batch_size=32, validation=(X_test, y_test),
        callbacks)
        PRINT "Training complete."
  Step 6. Evaluate Model
        PRINT "Evaluating..."
        EVALUATE on test set → accuracy, precision, recall, F1, AUC
        COMPUTE confusion_matrix, classification_report
        PRINT "Evaluation complete."
  Step 7. Visualize Results
        PRINT "Generating visualizations..."
        PLOT accuracy & loss curves
        PLOT confusion heatmap
        PLOT ROC & PR curves
        PLOT SHAP summary/beeswarm/dependence plots
        SAVE all figures
    Step 8. Predict Unlabeled Data
        PRINT "Predicting unlabeled..."
        LOAD & preprocess unlabeled inputs
        PREDICT with model
        SAVE predictions (CSV/JSON)
        PRINT "Prediction done."
  Step 9. Feature Importance (FI): PDP-ICE or SHAP, LIME
        PRINT "Computing Test..."
        INIT explainer = FI.KernelExplainer(model.predict, background_data)
        CALCULATE FI_values
        GENERATE FI plots (summary, waterfall, dependence)
        PRINT "FI analysis done."
END
```

Appendix B

The pseudocode for the adopted strategy of Pseudo-labeling and Fine-tuning during training is given below:

```
BEGIN
Step 1: Initialization
    LOAD pre-trained CNN model (trained on labelled images)
    FREEZE the convolutional blocks (low-level feature detectors - as required)
    SET fine-tuning parameters (learning rate, optimizer, loss function)
Step 2: Unlabeled Data Prediction
    FOR each sample in unlabeled dataset
      Do COMPUTE softmax probabilities using the CNN model
      IF maximum softmax probability ≥ 0.90
          Then ASSIGN provisional pseudo-label corresponding to highest probability
          class.
      ELSE
        DISCARD sample from pseudo-labeled dataset
      END
    END
Step 3: Pseudo-labeled Dataset Preparation
    MERGE original labeled data with pseudo-labeled data
    ASSIGN loss weights based on softmax probabilities:
      - IF softmax ≥ 0.95: loss_weight = 1.0 (high confidence)
      - IF 0.90 ≤ softmax < 0.95: loss_weight = scaled between 0.5–0.8 (moderate
    confidence)
Step 4: Secondary Fine-tuning Pass
    INITIALIZE loss function
    FOR each epoch DO
      TRAIN CNN model using combined dataset (labeled + pseudo-labeled)
      COMPUTE weighted loss using assigned loss weights
      VALIDATE model performance on separate validation set
      MONITOR validation loss
      IF validation loss improvement < 0.01% for 5 consecutive epochs THEN
        APPLY early stopping
        EXIT training loop
      END
    END
Step 5: Post-training Evaluation
    EVALUATE model robustness:
        CALCULATE accuracy, precision, recall, F1-score, and calibration error
        SELECT optimal confidence threshold (0.85) based on maximal F1-score for
        the unlabelled data evaluation
        GENERATE confusion matrices, ROC, and precision-recall curves for analysis
        PERFORM k-fold cross validation
        STORE final fine-tuned CNN model for future inference
END
```

Appendix C

The detailed parameter table summarizing the hyperparameters, execution durations, and processing specifics across all methodological steps is as shown below in Table 1.

Process Step	Key Parameters	Dimensions	Execution Time
Signal Acquisition	Accelerometer, Microphone	2 vibration signals (Vx, Vy), 1 audio signal (A)	Real-time (continuous acquisition)
Variational Mode Decomposition (VMD)	Number of IMFs: 8 (selected by Kurtosis criterion), Pearson threshold: RMS deviation	Input: 3 raw signals (time-domain), Output: denoised signals (3 × time-domain)	~100 ms per sample
Short-Time Fourier Transform (STFT)	Window length: 512 points, Overlap: 50 %, Sampling frequency: 50 kHz	Input: denoised signals (time-domain), Output: spectrogram (224 × 224 × 3 RGB)	~20 ms per signal

(*continued on next page*)

(*continued*)

Process Step	Key Parameters	Dimensions	Execution Time
Signal Fusion	Weighted Average Fusion (Optimal weights)	Input: 3 spectrograms (224 × 224 × 3), Output: fused spectrogram (224 × 224 × 3 RGB)	~15 ms per fusion
CNN Architecture	3 Conv. layers (64,128 filters, kernel: 3 × 3), 2 Max-pool (2 × 2), Dense layers (256 neurons), Dropout: 50 %, Activation: ReLU, Optimizer: Adam, Epochs: 500 (Early stopping: Patience = 5 epochs, Improvement <0.01 %)	Input: fused spectrogram (224 × 224 × 3), Output: Binary classification	~20 ms inference
Pseudo-Labeling (Only for training)	Softmax threshold: ≥0.90, Weight adjustment: ≥0.95 → 1.0, 0.90–0.95 → 0.5–0.8	Input: unlabelled samples, Output: pseudo-labelled samples	~10 ms/sample

Data availability

No data was used for the research described in the article.

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